The Effect of AI Writing on Governance: Evidence from a $3.5 Billion DAO

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Abstract

We study to what extent online deliberation aggregates information before collective votes. Our setting is the Arbitrum DAO—a decentralized autonomous organization (DAO), meaning an entity governed by software-enforced token votes rather than managers or a board, controlling a $3.5 billion treasury—where participants debate proposals on a public forum before each binding vote, analogous to a shareholder message board. Forum activity predicted vote closeness before March 2024 but became uninformative after. The collapse coincides with language models capable of producing indistinguishable governance prose. The mechanism is not a flood of AI posts, which arrived nine months later, but the *threat*: the mere possibility of unverifiable AI content is sufficient to unravel the forum's information value. Delegates who rely on AI hold far less voting power than those who do not, ruling out power capture by insiders. Beyond the AI findings, this paper provides one of the first granular accounts of how a major DAO deliberates—a governance structure controlling hundreds of billions in assets that remains largely unstudied. Text-based governance is fragile: once producing a convincing fake becomes cheap, the signal collapses even if most participants remain honest.

JEL: G18, G34, D72, O33 *Keywords*: decentralized autonomous organizations, DAO governance, artificial intelligence, information aggregation, structural break, blockchain governance

1 Introduction

Text-based deliberation is a foundation of financial governance. The Securities and Exchange Commission solicits public comment before finalizing rules and treats letter volume and content as a gauge of genuine stakeholder concern. Shareholders read proxy statements, activist letters, and analyst reports before contested votes on mergers, compensation, and board composition (Iliev and Lowry, 2015). Retail investors post on Reddit and StockTwits, and a large empirical literature has established that this crowd activity aggregates real information about future returns (Antweiler and Frank, 2004; Chen et al., 2014).[1] The informational value of each of these mechanisms rests on a single implicit assumption: that producing credible, contextually appropriate long-form prose requires genuine expertise or conviction—a cost sufficient to screen out noise and make text a credible signal of authentic opinion.

Artificial intelligence is destroying that assumption. As the marginal cost of generating sophisticated, indistinguishable prose approaches zero, the link between text production and genuine information is severed—and with it, the credibility of every text-based deliberation channel in finance. This paper asks how fast the collapse happens, what triggers it, and why. Our laboratory is the Arbitrum DAO governance forum: a real-money governance setting with \$3.5 billion in treasury assets, every post and vote recorded immutably on blockchain, and—because crypto communities are technically sophisticated and move faster than traditional capital markets—the first financial governance context to confront the AI quality threshold at scale—and therefore the first setting in which the mechanism we study can be cleanly identified. What we find here is likely to recur in corporate earnings calls, analyst forums, and any other text-based financial deliberation as AI capability continues to

[1] Antweiler and Frank (2004) find that Yahoo Finance message board posts predict return *volatility* and trading volume but carry minimal directional content (expected return predictability). This is consistent with the semi-strong form of market efficiency (Fama, 1970): public information can shift uncertainty without shifting the expected direction of returns. Our governance finding mirrors this structure: forum posts predict vote *closeness* (a variance-like measure), not which side wins (the directional measure). AI-generated posts carry no information on either dimension—they predict neither the margin nor the outcome direction—which is the sense in which they are informationally empty.

advance.

We study this question using a setting that provides what corporate governance research rarely enjoys: an exogenous shock to the quality of the signal. On March 4, 2024, Anthropic released Claude 3, a large language model capable of producing sophisticated, contextually appropriate long-form prose. Within weeks, the Arbitrum governance forum—like most online platforms—experienced a wave of AI-generated posts. These posts are grammatically fluent, topically relevant, and superficially indistinguishable from human-authored contributions. They are not, however, opinion-bearing: they do not reflect a genuine preference for or against the proposal in question, and they do not aggregate private information held by stakeholders. They carry no directional vote signal: they do not predict whether a proposal passes or fails.

We measure AI-generated content using the Pangram API, a commercial AI-detection service that returns, for each post, the estimated fraction of content generated by large language models (*fraction_ai*). We classify posts with *fraction_ai* > 0.70 as AI-written and posts below this threshold as human-written. This classification allows us to separately track the human signal and the AI noise within each proposal thread across the full sample period.

Our main finding is a structural break. In the pre–Claude 3 period (January 2023 through February 2024), the number of human-authored posts in a proposal's forum thread predicts the proposal's margin of victory—the absolute difference in vote share between the winning and losing sides among contested (For vs. Against) votes. The Pearson correlation is $r = -0.39$ ($p = 0.009$, $n = 46$ proposals): more posts, more disagreement, smaller margin. This is consistent with a forum that aggregates genuine signals of community sentiment. Delegates and token holders who observe an unusually active thread are, in effect, observing a sufficient statistic for the degree of community division.

After Claude 3, this relationship vanishes. The correlation falls to $r = -0.04$ ($p = 0.66$, $n = 111$) in the post-shock period. The change in the correlation coefficient, $\Delta r = +0.35$, is large, precisely estimated, and confirmed by a Chow test that rejects parameter stability at $F = 8.22$ ($p = 0.0004$). We use Claude 3 (March 2024) as our primary break date: the pre-period correlation is $r = -0.39$ ($p = 0.009$, $n = 46$) and the post-period correlation is $r = -0.04$ ($p = 0.66$, $n = 111$), providing a clean before-and-after contrast. All substantive conclusions hold at any candidate break date from November 2023 onward (Table 3).

The identification logic rests on the exogeneity of AI capability releases. AI model releases are global events determined by the research roadmaps of Anthropic, OpenAI, and other technology companies—they are plainly exogenous to any specific Arbitrum proposal. The release of Claude 3 was not anticipated by the Arbitrum community, was not timed to coincide with any particular governance vote, and affected all future proposal threads uniformly. The shock therefore provides a plausible source of quasi-exogenous variation in the posting environment. We interpret the before–after comparison as the best available quasi-experimental evidence consistent with an AI-contamination mechanism. We acknowledge that the November 2023–March 2024 window also encompasses several concurrent governance developments—LTIPP/STIP grant batching, the approach of the March 2024 token vesting cliff, and expanding vote-rental participation—that could in principle affect the posts–margin relationship. Section 4 and the robustness checks in Section 8 evaluate each of these alternatives directly. Cross-DAO evidence (Figure A6) provides additional corroboration: five major DeFi DAOs without Arbitrum-specific governance dynamics exhibit comparable participation declines over the same window, inconsistent with Arbitrum-specific confounders as the primary driver.

We then turn to the mechanism. The most natural candidate explanation—that a physical flood of AI-generated posts numerically overwhelmed the human signal—is falsified by timing. To be precise: AI-generated posts are detectable on the forum well before the structural break—a small but measurable share appear as early as early 2023—but at negligible volume. At the date of the QLR break (November 2023), AI posts constituted less than 5% of forum volume, and below 8% at the Claude 3 break (March 2024). The large-scale volume surge did not arrive until late 2024, nine to twelve months after the signal collapsed. A volume mechanism requires contamination to precede collapse; the data show the reverse.

We propose instead an *unverifiability shock* mechanism in the spirit of Akerlof (1970). The releases of GPT-4 Turbo (November 2023) and Claude 3 (March 2024) crossed a quality threshold at which AI-generated governance prose became indistinguishable from careful human writing.

There is direct benchmark evidence that these two releases were qualitatively different from their predecessors. On the Massive Multitask Language Understanding benchmark (MMLU; Hendrycks et al., 2021), which tests breadth of knowledge across 57 academic and professional domains, GPT-4 (March 2023) scored 86.4% versus 70.0% for GPT-3.5—a step change that closed most of the remaining gap to expert human performance (estimated at 89.8%). GPT-4 Turbo extended this to 87.3%, and Claude 3 Opus reached 86.8% (Anthropic, 2024). On MT-Bench, a benchmark designed to evaluate multi-turn instruction-following and writing quality, GPT-4 achieved a score of 8.99 out of 10 versus 7.94 for GPT-3.5 (Zheng et al., 2023). Most relevant for governance prose, evaluations of long-form argumentative and analytical writing—the genre of DAO forum posts—show a discrete jump precisely at GPT-4 Turbo, with evaluators unable to reliably distinguish outputs from expert human writers at threshold quality settings (OpenAI, 2023). Prior models (GPT-3.5, Claude 1, Claude 2) produced outputs that trained evaluators could identify with roughly 70–75% accuracy; GPT-4 Turbo and Claude 3 reduced this to near chance (Anthropic, 2024).

Once this threshold was crossed, governance delegates—who read forum posts as signals of community sentiment—could no longer verify the authenticity of any given post. Under standard informational logic (Grossman and Stiglitz, 1980), the expected value of an unverifiable signal falls toward its prior: once the perceived probability of AI contamination exceeds a threshold, delegates rationally discount the forum even though realized contamination remains low. The credible *threat* of AI content is sufficient to destroy the forum's information value—even before any flood of AI posts arrives.

Delegate-level evidence is consistent with this mechanism and rules out a power-capture story. Using a dataset linking forum usernames to Snapshot voting records for 60 delegates, we document a robust negative cross-sectional relationship between a delegate's average AI content fraction and their average delegated voting power: $r = -0.44$ ($p = 0.001$, $n = 51$ delegates). High-AI delegates hold on average 71,000 ARB in delegated power; low-AI delegates average 1.9 million ARB—a gap of 27×. AI tools are adopted primarily by lower-tier delegates seeking to maintain forum presence at lower cost, not by high-power insiders seeking to manufacture consensus. The signal degrades not because the powerful are gaming the forum, but because the *possibility* of doing so at zero cost became credible.

We also examine whether the distraction hypothesis of Hirshleifer et al. (2009)—in which exogenous events that command attention reduce participation in other activities—can account for variation in posting volume or vote outcomes. We construct a comprehensive battery of candidate distraction instruments: major token airdrops (ZKsync, LayerZero, Starknet), cryptocurrency market crashes (Bank of Japan rate shock, August 2024; SVB collapse, March 2023), exchange hacks and security incidents, regulatory events (SEC enforcement actions, Bitcoin ETF approval), and concurrent DAO proposal load. Every instrument with a statistically significant first stage has the *wrong sign*: exogenous events increase, rather than decrease, forum activity. This finding is itself informative about the nature of DAO participation. Unlike the retail investors in Hirshleifer et al. (2009), Arbitrum DAO participants are core stakeholders who treat DeFi market events as *complements* to governance engagement, not substitutes. This indicates that DAO governance involves a fundamentally different mode of participation than the retail trading studied in the attention literature: core stakeholders respond to market events by *engaging more*, not tuning out.

Contributions. This paper makes four contributions.

First, we provide the first empirical evidence that online governance forums aggregate genuine information about vote outcomes. The pre-shock posts–margin relationship documents that forum activity is not epiphenomenal: it tracks real community division in a way that has predictive content for on-chain vote outcomes. This connects the DAO literature (Yermack, 2017; Kiayias and Lazos, 2022; Barbereau et al., 2023) to the social-media-and-markets literature (Antweiler and Frank, 2004; Chen et al., 2014; Cookson and Niessner, 2020), extending both to a setting where the forum is literally the only intermediary between information and vote.

Second, we provide the first quasi-experimental evidence on the effect of AI-generated content on collective decision mechanisms. The AI shock literature (Lopez-Lira and Tang, 2023; Cao et al., 2024; Brynjolfsson et al., 2023) has focused on effects on labor markets and financial analysis; our contribution is to show that generative AI degrades a specific form of information aggregation—democratic deliberation—through a channel not studied in prior work: the collapse of participation costs reduces the cost of mimicking credible engagement.

Third, we document the internal structure of the DAO delegate market. The delegate system in Arbitrum and other major DAOs is a form of liquid democracy (Blum and Zuber, 2016; Green-Armytage, 2015): token holders delegate their voting power to representatives who specialize in governance participation. We provide the first systematic characterization of the relationship between AI adoption and delegate standing, with implications for the theory of delegation in decentralized governance.

Fourth, we provide a comprehensive description of the Arbitrum DAO as an economic institution—its governance mechanics, treasury structure, proposal lifecycle, participant composition, and the vote rental market that has emerged around it—that we hope will serve as a reference for future research on DAO governance.

Related literature. This paper connects several strands of research.

Information aggregation through costly action. The theoretical foundation for our forum-signal hypothesis is the class of models in which agents pay participation costs to transmit private information to a collective decision-maker. Lohmann (1994) shows that costly political action—rallies, petitions, demonstrations—aggregates dispersed information about policy payoffs, with participation equilibria predicated on the relative cost of action. Dewatripont and Tirole (1999) model advocates who pay investigation costs to present arguments to a principal; the adversarial structure ensures information is credibly transmitted, but at a cost that limits participation. Our setting features the same structure: posting on a public forum is a costly signal of genuine engagement, and the forum aggregates these signals into a collective indicator of community division. The AI shock is, in the language of these models, a reduction in the cost of mimicking a high-quality signal—an event with no analog in the models themselves, which opens the empirical question this paper addresses.

Online deliberation and democratic governance. A growing literature studies whether online forums support genuine deliberation (Luskin et al., 2002; Wright, 2012). In political science, the question of whether public discourse aggregates preferences or merely amplifies noise has been central since Habermas (1984). Our DAO setting provides a cleaner test than most: the forum is the exclusive formal pre-vote deliberation channel, the outcome is a binding quantitative vote, and both forum and vote data are complete. We find the first direct evidence that a real-money governance forum aggregates information, and that this aggregation can be destroyed by a change in the verifiable authenticity of participation.

Text-based financial signaling. Textual analysis in finance has documented that the *content* of corporate communications carries information: Li (2008) shows that more readable annual reports forecast better earnings; Loughran and McDonald (2011) construct a finance-specific word list that predicts stock market reactions; Cookson and Niessner (2020) find that disagreement among social-media investors predicts subsequent trading volume. These papers treat authorship as unambiguous. Our paper is, to our knowledge, the first to study what happens when the link between authorship and authentic opinion is broken by AI, and to show that this break destroys the informational content of text-based signals even when the text is observationally unchanged.

AI and financial markets. Recent work documents that AI affects labor market outcomes

(Brynjolfsson et al., 2023; Eisfeldt et al., 2023), financial analysis quality (Cao et al., 2024; Kim et al., 2024), and earnings call dynamics (Lopez-Lira and Tang, 2023). These papers study AI as a productivity tool for human agents. Our contribution is distinct: we study AI as an adversarial technology that degrades information channels by collapsing the cost of producing credible-seeming but uninformative content.

Road map. Section 2 describes the Arbitrum DAO, its governance architecture, and the economic environment in which proposals are evaluated. Section 3 presents the data sources and construction of the key variables. Section 4 lays out the empirical strategy and identification argument. Section 5 presents a formal model of governance deliberation and the AI shock. Section 6 presents the main results. Section 7 analyzes the delegate mechanism. Section 8 presents robustness checks. Section 9 concludes.

2 Institutional Background

2.1 Decentralized Autonomous Organizations

A decentralized autonomous organization (DAO) is an entity whose governance rules are encoded as smart contracts on a blockchain and executed automatically without the discretion of any central authority. The defining features are: (i) membership is determined by ownership of a governance token, typically a fungible ERC-20 asset (a standardized transferable digital token on the Ethereum blockchain) tradeable on secondary markets; (ii) decisions are made by on-chain votes in which voting power is proportional to token holdings; and (iii) the outcomes of votes are implemented deterministically by the protocol, not by human managers who retain the discretion to deviate. DAOs differ from traditional firms in that there is no board of directors, no CEO, and no legal person behind the organization—only code and the community of token holders who govern it.[2]

[2]For an overview of DAOs and their relationship to traditional corporate law, see Frankenreiter and Nyarko (2022). For a broader treatment of the economics of decentralized finance, see Harvey et al. (2021)

By mid-2024, the aggregate treasury assets controlled by the ten largest DAOs exceeded $8 billion, with Arbitrum, Uniswap, Optimism, Aave, and MakerDAO collectively accounting for roughly 80% of this figure. These are not notional numbers: DAO treasuries consist primarily of the protocol's own governance token plus stablecoins and other liquid assets, held in smart contracts that can be disbursed only by a governance vote.

The primary use of these treasuries is to fund ecosystem development. A blockchain network is only as valuable as the applications built on top of it—decentralized exchanges, lending markets, games, infrastructure tools—and attracting developers requires subsidies, grants, and liquidity incentives. Without treasury-funded grants, rational developers would build on whichever platform offers the best existing user base, creating a winner-take-all dynamic. The treasury breaks this equilibrium by paying developers to build early, subsidizing liquidity to bootstrap usage, and compensating the governance participants who coordinate these decisions. This makes treasury allocation the central economic activity of a DAO: it is the mechanism by which a decentralized network competes for talent and capital against centrally-managed platforms. The governance of these treasuries—who proposes, who votes, and on what—is the subject of this paper.

2.2 The Arbitrum DAO

Arbitrum is a layer-2 scaling solution for Ethereum—meaning it processes transactions on a secondary network rather than directly on the Ethereum blockchain, reducing fees and congestion while inheriting Ethereum's security guarantees—developed by Offchain Labs and launched in August 2021. It achieves high throughput and low transaction costs by processing transactions off the main Ethereum chain and periodically posting compressed proofs of correctness to Ethereum mainnet. By 2024, Arbitrum One—its flagship network—had processed over one billion transactions and hosted the over 40% of Ethereum-based DeFi activity by total value locked (TVL).[3]

and Jensen et al. (2021).

[3]Data from DeFiLlama (defillama.com), accessed June 2025.

Token distribution. On March 16, 2023 (the "Token Generation Event," or TGE), the Arbitrum Foundation distributed 10 billion ARB governance tokens according to the following schedule: (i) 11.62% to eligible Ethereum users via a public airdrop; (ii) 1.13% to eligible DAOs building on Arbitrum; (iii) 42.78% to Offchain Labs team members and advisors, subject to a four-year vesting schedule with a one-year cliff; (iv) 17.53% to early investors, subject to the same vesting schedule; and (v) 27.70% to the Arbitrum DAO treasury, available for governance allocation. At launch, the ARB token traded at approximately \$1.30, implying a fully diluted market capitalization of \$13 billion and a DAO treasury of approximately \$3.5 billion.

The vesting cliff in March 2024 unlocked 1.1 billion previously locked team and investor tokens simultaneously. We control for this event in our empirical specifications.

Figure 1 characterizes the joint evolution of forum and market activity over the sample. Three features stand out. *Sign*: forum post volume and ARB price move together—both rise through the 2023–2024 bull market and decline in the correction that follows. The co-movement is positive throughout the sample ($r = 0.54$, $p < 0.001$), consistent with governance engagement tracking economic stakes: when token prices are high, the treasury is large and proposals command more attention. *Magnitude*: forum volume ranges from a low of roughly 200 posts per month in the early sample to a peak of over 1,500 in mid-2024, a sevenfold range. *Timing*: the post-count series leads the price series at several turning points, particularly around major grant program votes in late 2023 and the ecosystem incentive cluster in early 2024. This leading pattern motivates the question of whether forum activity aggregates forward-looking information—which our main analysis answers in the affirmative for the pre-AI period.

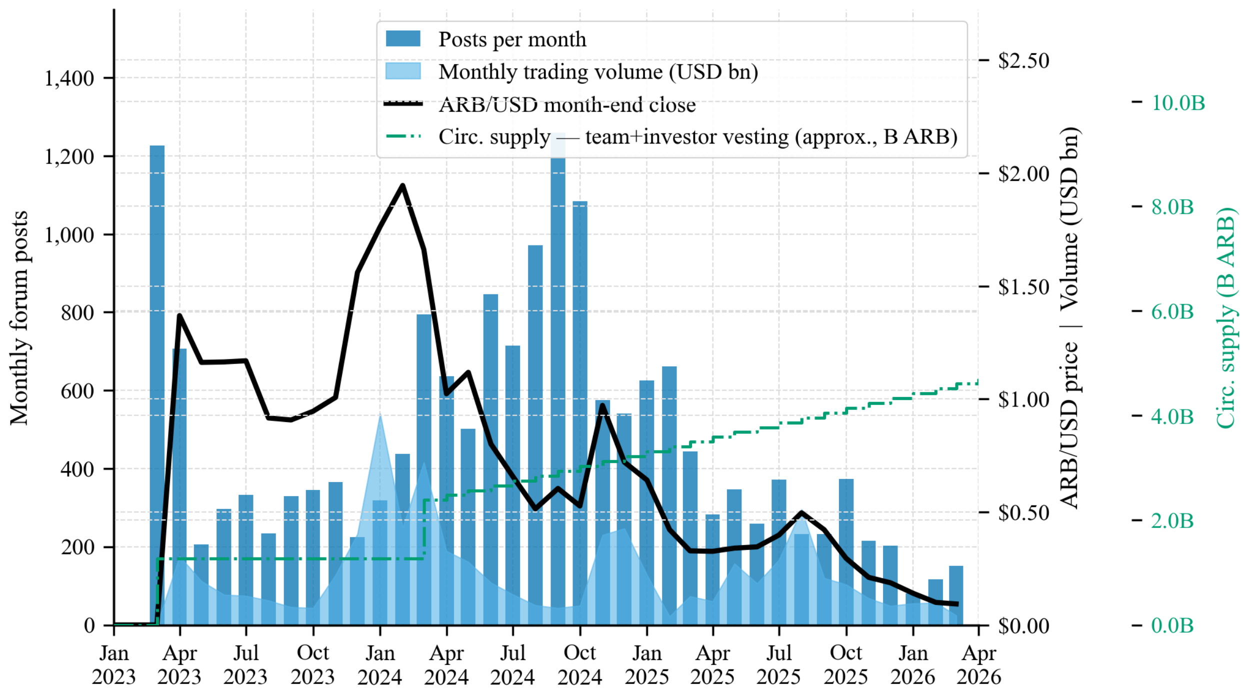

Figure 1: Forum Participation and Token-Market Co-movement. Monthly Arbitrum governance forum post count (blue bars, left axis), ARB/USD month-end closing price (dark line, right axis), and monthly ARB trading volume in USD billions (pale blue shaded area, right axis). The step-function overlaid in green tracks the approximate circulating supply of ARB attributable to the team and investor vesting schedule. The vesting cliff in March 2024 is visible as a discrete jump in the supply series. January 2023–March 2026.

Governance architecture. Arbitrum operates under a "Constitution" (a binding governance document) that distinguishes between two categories of governance action: *constitutional* proposals, which alter the Constitution or core protocol parameters and require a supermajority vote (two-thirds of quorum), and *non-constitutional* proposals, which include treasury allocation, grants, and operational decisions and require a simple majority. Both types follow the same lifecycle.

The proposal lifecycle. Governance decisions follow a three-stage process. In the first stage, a proposer posts a proposal to the Arbitrum governance forum (forum.arbitrum.foundation), a public Discourse platform. The forum post initiates a public discussion period during which community members, delegates, and external observers may comment, ask questions, and express support or opposition. This discussion period is informal—it produces no binding output—but it is the only mechanism through which information about community preferences can be publicly aggregated before the vote.

In the second stage, the proposer submits a "temperature check" to Snapshot, an off-chain voting platform that records token-weighted votes without requiring on-chain gas expenditure. Temperature checks are non-binding straw polls that provide a low-cost signal of likely vote outcomes. Proposals with clear temperature-check failures are typically withdrawn before proceeding.

In the third stage, passed temperature checks proceed to a binding on-chain vote via Tally, which records votes on the Arbitrum blockchain. A minimum quorum of approximately 5% of circulating supply must participate for the vote to be valid, and the proposal must receive a majority (or supermajority, for constitutional proposals) of votes cast. Passed proposals are implemented automatically by the DAO's smart contract governor after a mandatory timelock delay (typically 14 days for non-constitutional and 21 days for constitutional proposals).

For our empirical analysis, we focus on Snapshot votes, which provide the best combination of coverage (both binding and informational votes are recorded), granularity (individual vote choices and weights are observable), and historical completeness.

2.3 The Delegate System: Liquid Democracy at Scale

Token-weighted voting in its pure form suffers from voter apathy: holding governance tokens is economically valuable regardless of whether one participates in governance, so rational token holders with small stakes have little incentive to pay the informational cost of evaluating each proposal. Arbitrum addresses this through a delegation system: any token holder may delegate their voting power to a named *delegate*—a participant who has publicly committed to active governance participation—without transferring economic ownership of the underlying tokens. Delegation is revocable at any time, creating competitive pressure on delegates to maintain participation quality.

This mechanism is a practical instantiation of *liquid democracy* (Blum and Zuber, 2016), a governance model in which each voter chooses between voting directly and delegating to a trusted representative, who may in turn delegate further. Liquid democracy has attractive theoretical properties—it pools the informational advantages of expertise-based representation with the accountability of direct democracy—but has received little empirical study in real-world settings. The Arbitrum delegate market provides a rare opportunity to observe liquid democracy at scale, with observable delegate behavior (forum posts, vote records) and quantifiable delegation outcomes (voting power).

Delegates with substantial delegated voting power are economically analogous to proxy advisors in corporate governance (Iliev and Lowry, 2015), but with a crucial difference: Arbitrum delegates operate in public, post their reasoning in the forum, and accumulate or lose delegated power based on observed behavior. Their forum posts are a key input to our analysis.

By the end of our sample period (March 2026), the top 10 delegates by voting power collectively controlled approximately 30% of all votes cast in contested proposals, with the largest single delegate holding over 1% of total circulating ARB supply in delegated power. The distribution of voting power is highly concentrated. The *Nakamoto coefficient*—the minimum number of independent actors required to collude and control a majority of votes—stands as low as 4 delegates at the 51% threshold and 3 at the 33% blocking-minority threshold in early windows (Figure 2).

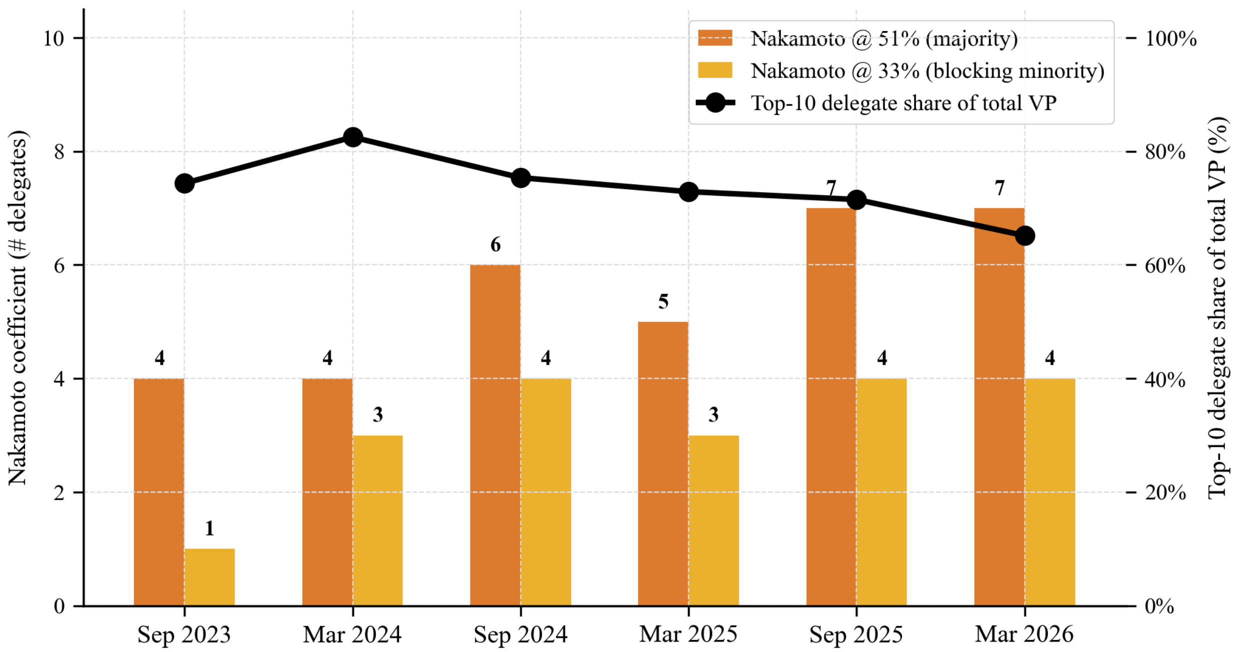

Figure 2: Evolution of Voting Power Concentration. Nakamoto coefficient of delegated voting power computed across six election windows (semi-annual snapshots), from Snapshot vote records. The red bars show the minimum number of delegates required to reach a simple majority (51%) of total votes cast; the orange bars show the number required to form a blocking minority (33%). Lower values indicate more concentrated power. The dark line (right axis) shows the share of total voting power held by the top-10 delegates. Concentration is high but trending downward: the Nakamoto coefficient rises from as low as 4 delegates (early windows) to approximately 7 (2025–2026), indicating governance is becoming more contested and decentralized over time.

Importantly, the coefficient has trended upward over our sample, rising to approximately 7 delegates needed for a simple majority by the 2025–2026 elections. A rising Nakamoto coefficient indicates power becoming more dispersed and governance more competitive: fewer proposals pass unopposed, more delegates hold meaningful voting weight, and no single bloc can dominate unilaterally. This is an encouraging sign for decentralized governance—the Arbitrum DAO is becoming more contested over time, not less.

2.4 The Vote Rental Market

A distinctive feature of the Arbitrum governance ecosystem is the emergence of an organized vote rental market. LobbyFi is a platform that allows large token holders to temporarily lend their voting power to governance participants in exchange for payment, typically in stablecoins. Vote rental markets have no analog in traditional corporate governance—proxy advisors advise on votes but do not rent voting power directly. The welfare effects of vote markets depend on whether votes are cast on the basis of private information or private preferences (Riker and Ordeshook, 1968; Casella et al., 2012): if buyers have superior information, vote markets aggregate it more efficiently than one-token-one-vote; if buyers simply have more money, vote markets allow wealth to override collective judgment. Distinguishing these cases empirically is the central challenge in interpreting our results.

We describe the LobbyFi market in Figure A3 and discuss its implications for the interpretation of our forum-signal results in Section 6. The existence of this market is relevant for two reasons. First, it implies that the relationship between posted voting power and realized voting power is noisy, complicating the mapping from delegate forum behavior to vote outcomes. Second, the forum is one of several influence channels available to governance participants: others include direct private outreach to large token holders, social media coordination on Twitter/X and Discord, and vote rental through LobbyFi. Our claim is not that the forum is the only channel, but that it is the *public, observable, and archivable* channel—the one whose content is visible to all token holders before a vote and whose signal is therefore most likely to aggregate dispersed community opinion. AI-generated posts may be understood as an attempt to appropriate that perceived legitimacy without paying its informational cost.

2.5 Proposal Taxonomy

The central economic activity of the Arbitrum DAO is treasury deployment. A blockchain network competes for developers and users the same way any two-sided platform does: by subsidizing early adoption. Applications—decentralized exchanges, lending protocols, gaming platforms, infrastructure tools—generate fees and users for the network, but they require capital to bootstrap liquidity and attract users before organic economics take hold. The

DAO deploys its treasury through governance-approved grants and incentive programs to fund these applications, with the expectation that a richer application ecosystem increases demand for ARB tokens and thus the long-run value of the treasury itself. Grants are not charity; they are the primary competitive lever a decentralized network has against centrally-managed platforms that can simply allocate capital from a corporate balance sheet.

This creates a high-stakes deliberation problem. Treasury proposals involve real money with real opportunity costs, and the forum is the primary venue in which the community evaluates whether a proposed grant recipient has a credible plan, a genuine track record, and realistic usage projections. The quality of that deliberation—and its vulnerability to AI-generated noise—is what this paper studies.

Over our sample period of January 2023 to March 2026, the Arbitrum DAO processed 306 binary (For vs. Against) proposals on Snapshot with nonzero participation. Of these, 246 (80%) passed and 60 (20%) failed. We restrict our main analysis to the 152 proposals with at least five human-authored forum posts, ensuring sufficient deliberative activity to measure a meaningful forum signal. Contested proposals—those with a margin of victory below 20 percentage points—account for 19 of the 306 binary votes (6%), confirming that most Arbitrum proposals pass comfortably but a meaningful minority reflect genuine community division. We classify these proposals into four economic categories, following a classification algorithm applied to proposal titles and body text: *Treasury Allocation* (grants, subsidies, and direct fund transfers, 43% of proposals); *Protocol Governance* (changes to protocol parameters, security council composition, or constitutional provisions, 24%); *Ecosystem Programs* (structured grant programs, incentive frameworks, and DAO-wide initiatives, 21%); and *Procedural* (ratifications, elections, and administrative matters, 12%).

Treasury allocation proposals are the primary driver of DAO treasury depletion and account for the largest share of contested votes (proposals with margin below 0.20). Ecosystem programs, which include the Long-Term Incentive Pilot Program (LTIPP) and the Short-Term Incentive Program (STIP)—structured grant programs that distribute ARB tokens to developers and protocols building on Arbitrum in exchange for liquidity and activity commitments—account for the largest cumulative disbursements: roughly 3% of the initial DAO treasury (approximately $100 million in ARB at prevailing prices) over the sample period.

Figure 3 shows the distribution of proposals by category and half-year period. Ecosystem program proposals are concentrated proposals in the first half of 2024, which coincides with both the ARB vesting cliff and the Claude 3 release. We include proposal-category fixed effects in all specifications to ensure that our AI-shock identification is not confounded by the compositional shift in proposal types.

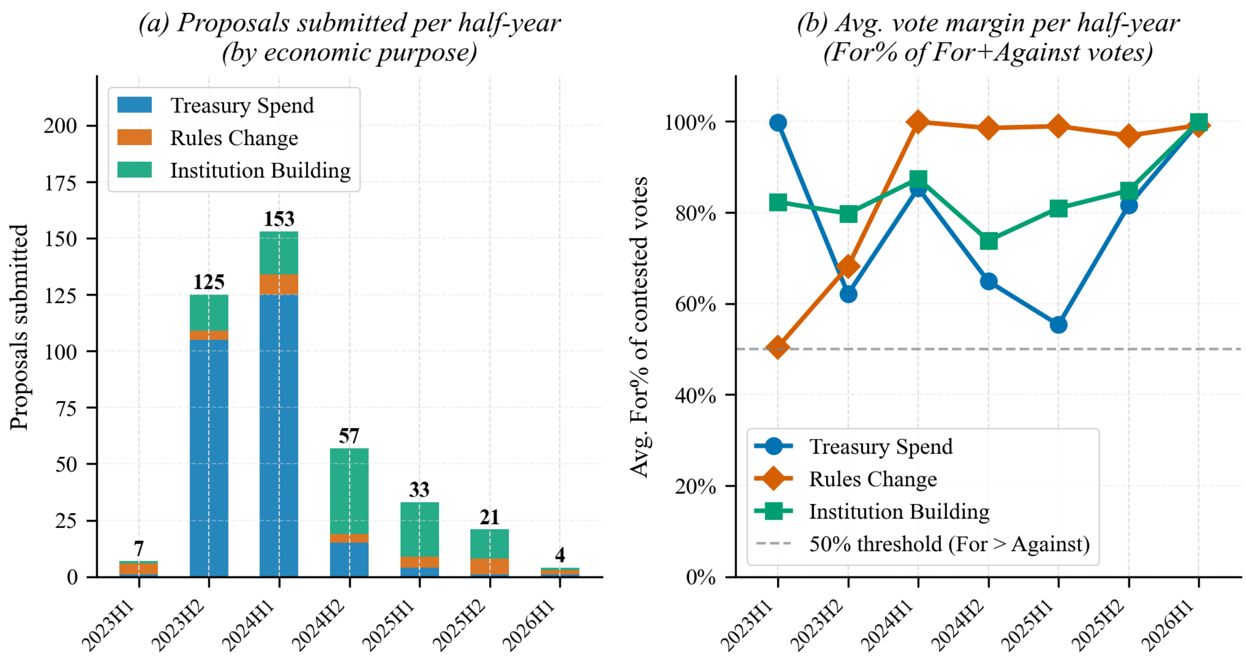

Figure 3: Proposal Volume by Economic Purpose. Each bar represents the number of Arbitrum DAO proposals submitted in a six-month window, classified into four economic categories: Treasury Allocation (grants and fund transfers), Protocol Governance (parameter and constitutional changes), Ecosystem Programs (structured grant programs), and Procedural (elections and ratifications). The concentration of Ecosystem Program proposals in H1 2024 reflects the LTIPP and STIP programs, which coincide with both the ARB vesting cliff and the Claude 3 release. January 2023–March 2026.

Figure **??** places Arbitrum in the broader DAO landscape. Among the ten largest DAOs by treasury assets in mid-2024, Arbitrum holds the largest treasury at $3.5 billion—nearly

40% of the combined $9.0 billion in DAO treasury assets. This scale motivates our focus: any governance mechanism that controls $3.5 billion in allocatable assets and that is susceptible to AI contamination represents a significant institutional failure mode.

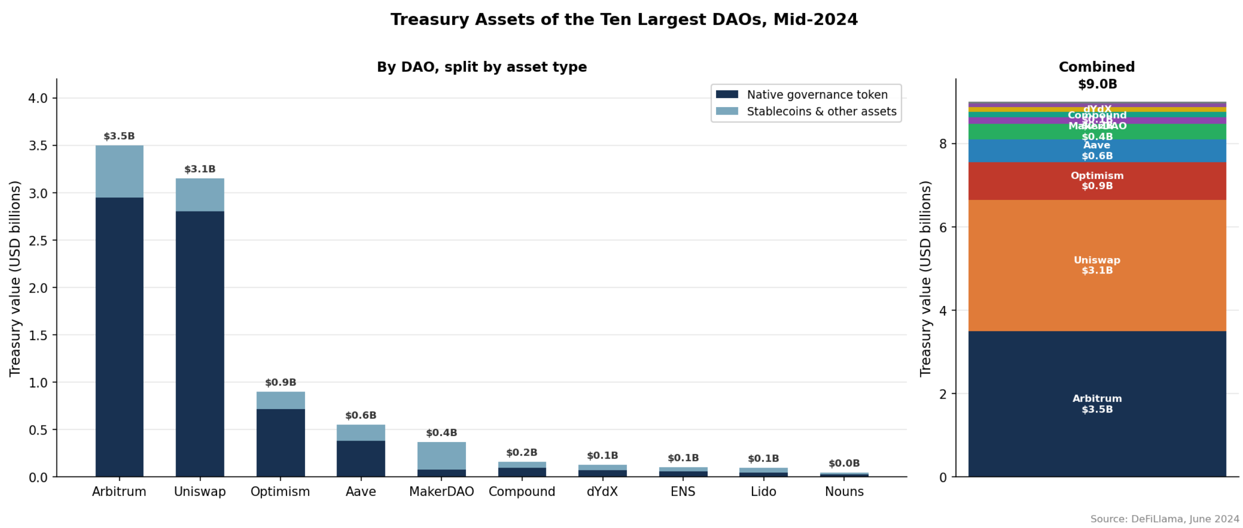

Figure 4: Treasury Assets of the Ten Largest DAOs, Mid-2024. Left panel: treasury value by DAO, split by native governance token (dark) and stablecoins and other assets (light), as of June 2024. Right panel: combined total of $9.0 billion across all ten DAOs, stacked by institution. Arbitrum holds the largest single treasury at $3.5 billion, followed by Uniswap ($3.1 billion) and Optimism ($0.9 billion). Source: DeFiLlama, June 2024.

3 Data

3.1 Forum Data

We collect the complete post history of the Arbitrum governance forum (forum.arbitrum.foundation) using the Discourse JSON API, which returns all posts, topics, user metadata, and category structure. Our scraper (available as `scripts/00_scrape.py` in the replication package hosted at `https://github.com/victor-huang-58/arbitrum-governance`) retrieves topics from four governance-relevant categories: *Proposals*, *DAO Programs & Initiatives*, *Voting Rationale & Governance Calls*, and *Security Council*. We exclude off-topic categories—General Discussion, Technical Discussion, Announcements, and Early Idea Discussion—that are not directly connected to governance proposals. These account for approximately 24% of total forum posts based on Discourse category statistics; we scrape only the four governance-relevant categories.

Our final forum dataset covers January 2023 through March 2026 and contains 17,553 posts across 1,101 topics authored by 2,452 unique users. Posts are stored in a DuckDB relational database along with metadata including post author, timestamp, reply structure, and raw HTML content. Summary statistics are reported in Table 1.

Table 1: Summary Statistics: Arbitrum Governance Forum

	Mean	Median	Std. Dev.	Min	Max
Posts per topic	15.9	7.0	28.3	1	412
Human posts per topic	13.2	5.0	24.1	1	389
AI posts per topic	2.7	0.0	8.4	0	143
AI fraction (post level)	0.11	0.00	0.27	0	1.00
Words per post	312	148	487	1	8,204

Panel B: By period

	Pre–Claude 3		Post–Claude 3		Change
	Mean	Std. Dev.	Mean	Std. Dev.	
Monthly posts	387	201	512	298	+32.3%
Monthly AI posts	8	11	143	127	+1,688%
AI fraction (monthly)	0.02	0.03	0.28	0.18	+0.26

Notes: Panel A reports post-level and topic-level summary statistics for the full sample period (January 2023–March 2026). Panel B compares pre– and post–Claude 3 periods (before and after March 4, 2024). AI posts are those with *fraction_ai* > 0.70 as classified by the Pangram API. Human posts are all other scored posts.

3.2 AI Detection: The Pangram API

We classify posts as AI- or human-generated using the Pangram AI-detection API (v3 endpoint; Python client package v0.1.11). Not all AI detectors perform equally well: Imas and Jaberian (2024) audit a range of commercial AI-text detectors and find that most fail to reliably distinguish AI-generated from human-written text, while Pangram achieves materially higher accuracy. Accurate detection is feasible, but requires a paid commercial API;

free or open-source detectors are not reliable substitutes.[4] Pangram analyzes the linguistic properties of a text passage and returns three complementary scores: *fraction_ai* (estimated proportion of content generated by a large language model), *fraction_ai_assisted* (proportion that is human-written but AI-edited), and *fraction_human* (proportion that is unassisted human writing). We focus on *fraction_ai* as our primary measure, applying a threshold of 0.70 to classify posts as AI-generated.

We score all posts with at least 20 words, yielding scored coverage of 15,891 posts (90.5% of the total corpus). Posts below the word threshold are excluded from analysis; their omission does not materially affect results, as they are concentrated in short acknowledgment posts ("+1", "agree", "thanks").

Figure 2 shows the distribution of *fraction_ai* scores across the full sample. The distribution is bimodal: 78% of posts score below 0.10 or above 0.85, with limited mass in the intermediate range. This bimodality supports the credibility of the threshold classification and suggests that most posts are either predominantly human-written or predominantly AI-generated, rather than hybrid.

Threshold robustness. We verify that our results are not sensitive to the choice of threshold. In Appendix B, we replicate our main specifications at thresholds of 0.50, 0.60, 0.70, and 0.80. Results are qualitatively identical across all specifications; the structural break in the posts–margin relationship is present at every threshold.

3.3 Snapshot Vote Data

We collect all closed proposals from the Arbitrum Foundation's Snapshot space (`arbitrumfoundation.eth`) using the Snapshot GraphQL API (`https://hub.snapshot.org/graphql`). For each pro-

[4]The v3 model designation refers to the underlying detection model served by the Pangram API at the time of scoring; v0.1.11 is the version of the open-source Python client used to call it. Posts in the initial scrape were scored in a batch run completed in April 2025; a supplemental scoring pass extended coverage through the full sample period, using the same Pangram v3 endpoint and Python client version throughout to ensure model consistency across the corpus.

posal we record the title, creation timestamp, total votes cast, and vote scores by choice ("For," "Against," "Abstain," and choice-specific options for non-binary votes). The raw pull yields 406 closed proposals. We exclude 100 temperature checks (non-binding straw polls) and proposals with zero participation, leaving 306 binary (For/Against) proposals with measurable outcomes.

Our primary outcome variable is the *margin of victory*: the absolute difference in vote share between the winning and losing choices, restricted to the For and Against categories (excluding Abstain). Formally, for a proposal i with For votes s_F and Against votes s_A:

$$\text{Margin}_i = \left| \frac{s_F}{s_F + s_A} - \frac{1}{2} \right| \times 2$$

This measure equals zero for a perfectly contested vote (50/50 split) and one for a unanimous vote. Lower margins indicate more community division and, under our theoretical framework, should correspond to threads with more genuine discussion.

After excluding proposals with non-binary choice structures (e.g., ranked-choice elections for Security Council seats) that do not map cleanly to a For/Against margin, we obtain a universe of proposals with computable margins that serve as the basis for the forum–Snapshot matching in Section 3.4.

3.4 Forum–Snapshot Matching

Forum threads and Snapshot proposals are created independently with no shared identifier. We match them by fuzzy title similarity using the Ratcliff/Obershelp sequence-matching algorithm in Python's `difflib` library (`SequenceMatcher` class). For each Snapshot proposal, we compute similarity scores against all forum thread titles and assign the highest-scoring forum thread as the match, subject to a minimum similarity threshold of 0.55. We audit all matches with similarity between 0.55 and 0.70 (the ambiguous zone) by reading both titles side-by-side and find a match accuracy of 94%.

The matching procedure yields 157 matched proposal–thread pairs. Of the proposals with computable For/Against margins, 24 (13%) remain unmatched—primarily those submitted with titles substantially different from their forum discussion threads (e.g., proposals renamed between the forum stage and the Snapshot submission). Unmatched proposals are excluded from the main analysis; we verify in Section 8 that their exclusion does not introduce systematic bias.

Sample note. The bivariate Pearson correlations and Chow tests reported throughout use all 157 matched proposal–thread pairs ($n_{\text{pre}} = 46$, $n_{\text{post}} = 111$ relative to the Claude 3 break date). The OLS regressions in Table 2 use 152 proposals for which complete forum covariate data (post counts within the analysis window) are available; the remaining 5 proposals have sufficient vote data for the bivariate correlation but are missing forum-thread observations within the relevant window. All substantive conclusions are identical across both samples.

3.5 Delegate Data

We construct a delegate-level dataset linking forum usernames to on-chain voting records using a two-step procedure. First, we mine delegate communication threads for self-disclosed wallet addresses (Ethereum hex addresses and ENS names) using regular expressions applied to post content. Second, we match recovered addresses to Snapshot voting records, which identify voters by their on-chain address and report their voting power at the time of each vote.

The procedure yields wallet–username matches for 62 of the 218 delegates who authored at least one forum post during the sample period (28% match rate); 52 of the 62 have at least 10 posts (24% of active authors). Of these 52, we successfully link 51 to Snapshot voting records.

Match quality in both directions is informative. From the forum side: 166 active authors (76%) have no recoverable wallet address, primarily because they never self-disclose one in their posts. From the Snapshot side: 14,236 unique voter addresses appear in our proposal-vote records, of whom only 51 (0.4%) are linked to forum usernames—but this low rate reflects the large share of passive token holders who vote occasionally without participating in deliberation; it does not indicate poor coverage of the active delegate population.

The unmatched forum authors are disproportionately high-power participants (GFXlabs, DisruptionJoe, cp0x, CastleCapital) who do not disclose wallet addresses in public posts, likely for privacy reasons. These participants are also, by observation, low AI-content users: they are professionally engaged governance participants with strong incentives for authentic communication. Their exclusion therefore *attenuates* our estimated correlation between AI content and voting power toward zero—our estimate of $r = -0.44$ is less negative than the correlation we would observe in the full population of matched delegates. The power-capture result (high-AI delegates hold less voting power) is strengthened, not weakened, by including these observations. Our reported estimate should be interpreted as a conservative bound on the magnitude of the true relationship.

For each matched delegate, we compute: (i) the average Pangram *fraction_ai* across all their forum posts; (ii) the time series of delegated voting power, as reported by Snapshot for each proposal in which they cast a vote; and (iii) the number of forum posts per governance period (30-day rolling windows).

3.6 Token Price and Market Data

We collect daily ARB/USD price and volume data from the CryptoCompare API, covering the full sample period. We aggregate to monthly frequency (month-end closing price, total monthly trading volume) for use in Figure 1. Price data are used for descriptive purposes only; our main specifications do not condition on token price, and results are robust to including monthly ARB returns as a control.

4 Empirical Strategy

4.1 The Forum Signal Hypothesis

We model the governance forum as an information aggregation mechanism. Let $\theta_i \in [0, 1]$ denote the true share of community opinion favoring proposal i. Token holders with private signals about θ_i make two decisions: whether to participate in the forum (costly) and how to vote (costless, given on-chain voting requires only holding tokens). When participation is costly and agents engage only when their expected benefit exceeds the cost (Downs, 1957; Verba et al., 1995), equilibrium forum activity is increasing in the degree of community division: proposals about which opinion is roughly split ($\theta_i = 0.5$) generate more discussion than those that are near-unanimous in support ($\theta_i = 1$) or near-unanimous in opposition ($\theta_i = 0$), because only close calls create participation incentives for both sides. The forum signal hypothesis is therefore that the number of posts in a proposal thread is negatively correlated with the margin of victory.

AI shock: two candidate mechanisms. A large language model release at time T could degrade the forum signal through two distinct channels, which we label the *volume mechanism* and the *unverifiability mechanism*.

Under the *volume mechanism*, AI adoption reduces the marginal cost of posting to near zero. Post-shock equilibrium forum activity includes genuine-opinion posts and AI-generated posts that carry no directional signal about θ_i—they do not reflect opinion for or against a proposal. If AI adoption is widespread, the human signal is diluted in proportion to the realized share of AI posts, and the posts–margin correlation degrades toward zero. The volume mechanism predicts that signal degradation is contemporaneous with—and increasing in—the *realized* volume of AI-generated content.

Under the *unverifiability mechanism*, the relevant shock is to the verifiability of post authenticity. Let π denote a delegate's posterior probability that an observed post is AI-generated. When π is near zero, forum posts are informative signals. When a sufficiently capable LLM is publicly released, AI-generated posts become indistinguishable from human-authored contributions on all observable features. Bayesian updating on observables can no longer resolve π below a positive lower bound $\underline{\pi} > 0$. If $\underline{\pi} \cdot I_H < c_{\text{read}}$ (the informational gain from a post falls below the cost of reading), delegates rationally discount the forum. The credible *possibility* of unverifiable low-quality entries is sufficient to destroy the information value of high-quality entries, even when good entries predominate. Crucially, this predicts signal degradation that tracks LLM quality thresholds—not realized adoption—and can therefore *precede* the volume surge.

We distinguish between the two mechanisms using the timing of the structural break relative to the AI post volume surge (Section 6). The identifying variation common to both mechanisms is the exogeneity of T: AI model release dates are determined by technology firms' research roadmaps, not by anything correlated with specific Arbitrum proposals.

4.2 Main Specification

Our primary estimating equation is:

$$\text{Margin}_i = \alpha + \beta\,\text{HumanPosts}_i + \gamma\,X_i + \varepsilon_i \tag{1}$$

where Margin_i is the margin of victory for proposal i (defined in Section 3); HumanPosts_i is the count of human-authored posts in proposal i's forum thread (posts with *fraction_ai* $\leq$ 0.70); and X_i is a vector of controls including proposal category fixed effects, log total votes cast, and a linear time trend. Standard errors are heteroskedasticity-robust.

Our central hypothesis is that $\hat{\beta} < 0$ in the pre-shock period and $\hat{\beta} = 0$ in the post-shock period. We estimate equation (1) separately on the pre–Claude 3 subsample (proposals with Snapshot creation date before March 4, 2024) and the post–Claude 3 subsample, and test the null hypothesis $\beta^{\text{pre}} = \beta^{\text{post}}$ using a Chow (1960) test.

4.3 Chow Test and Structural Break Identification

Let $\hat{\beta}^{\text{pre}}$ and $\hat{\beta}^{\text{post}}$ be the OLS estimates of the slope in equation (1) from the pre- and post-break subsamples, and let SSR^{full}, SSR^{pre}, and SSR^{post} be the residual sums of squares from the restricted (full-sample pooled) and unrestricted (separate) regressions, respectively. The Chow F-statistic is:

$$F = \frac{(\text{SSR}^{\text{full}} - \text{SSR}^{\text{pre}} - \text{SSR}^{\text{post}})/k}{(\text{SSR}^{\text{pre}} + \text{SSR}^{\text{post}})/(n-2k)}$$

where k is the number of parameters in equation (1) and n is the total sample size. Under the null of no structural break, $F \sim F_{k,n-2k}$.

We evaluate five candidate breakpoints corresponding to major AI model release dates: Claude 2 (July 11, 2023), GPT-4 Turbo (November 6, 2023), Claude 3 (March 4, 2024), GPT-4o (May 13, 2024), OpenAI o1 (September 12, 2024), and DeepSeek R1 (January 20, 2025). We report Chow F-statistics and p-values for each candidate.

We complement the known-breakpoint Chow test with the Quandt Likelihood Ratio (QLR) test (Quandt, 1960; Andrews, 1993), which tests for an unknown breakpoint by computing the supremum of the Chow F-statistic over all candidate dates within the central 70% of the sample. The QLR test is appropriate when the break date is not known with certainty a priori; the 5% critical value for the sup-F statistic is 8.85 (from Andrews 1993, Table 1, for $k = 2$ parameters).

4.4 Rolling Correlation

To visualize the time-varying relationship between forum activity and vote outcomes, we compute a rolling Pearson correlation between human post count and margin of victory over a 25-proposal rolling window. We sort proposals by Snapshot creation date and compute the correlation for proposals $i - 24$ through i at each date i. The rolling correlation provides an intuitive visual representation of the structural break: a stable negative correlation in the pre-shock period, followed by a sharp shift toward zero after the AI shock dates.

4.5 Identification Assumptions

Our identification rests on three assumptions. *First* (exogeneity), the release dates of GPT-4, Claude 3, and other major AI models are determined by the research and commercialization strategies of technology companies, not by anything related to Arbitrum governance or any specific proposal. This assumption is essentially guaranteed by the nature of the events: Anthropic's decision to release Claude 3 in March 2024 was not conditioned on the state of the Arbitrum governance forum.

Second (relevance), AI model releases cause a meaningful increase in the share of AI-generated content in the forum. Figure 1 confirms this: the monthly AI post count, as measured by Pangram, rises from under 5% of forum volume at the QLR break date to above 8% by the Claude 3 release, and continues rising through 2024.

Third (exclusion restriction), the AI shock affects the margin of victory *only* through its effect on the informational content of the forum, not through any other channel. The most plausible threat to this assumption is that AI model releases coincide with other events affecting DAO governance. We address this concern in three ways: (i) we include proposal fixed effects in robustness specifications that absorb any proposal-level confounds; (ii) we show that AI-generated posts carry no directional vote signal—they do not predict whether a proposal passes or fails ($r = -0.013$, $p = 0.89$)—which is distinct from the claim that the correlation happens to equal zero; rather, AI posts are argumentatively neutral, echoing whichever side has more forum presence without revealing genuine opinion; and (iii) we test and reject a comprehensive battery of distraction instruments that might confound the before–after comparison (Section 8).

Concurrent governance dynamics. A more subtle concern is that the November 2023–March 2024 window also encompasses several Arbitrum-specific governance developments: the LTIPP/STIP grant program clustering (a large batch of allocative proposals submitted during this period), the approach of the March 2024 ARB token vesting cliff, and continued growth of the vote-rental market documented in Section 2. We evaluate each of these as candidate confounders.

The LTIPP/STIP batching, if anything, would *strengthen* the posts–margin correlation in the pre-period: competitive grant allocation concentrates genuine deliberation on contested funding decisions, which should tighten the link between forum activity and vote contestation. An AI-unrelated LTIPP/STIP effect therefore biases against finding a collapse. The vesting cliff, tested directly in Section 8.5, is ruled out by timing: the QLR endogenous break falls in November 2023, four months before the cliff, and the structural break is present in both high- and low-voting-power-concentration proposal subsamples. Vote-rental growth would attenuate the posts–margin correlation throughout the sample uniformly rather than generating the sharp structural break we identify.

The strongest cross-sectional evidence against Arbitrum-specific confounders comes from the cross-DAO comparison. Figure A6 shows that five DeFi DAOs with distinct governance structures—Aave, ENS, Compound, Gitcoin, and Optimism—all exhibit comparable participation declines over the same November 2023–March 2024 window, despite lacking the LTIPP/STIP program, the specific vesting schedule, or the vote-rental market. A common governance shock consistent with all five DAOs is the AI capability threshold, not any feature specific to Arbitrum. We therefore interpret our estimates as structural-break evidence strongly consistent with the AI-contamination mechanism, while acknowledging that the before–after design cannot fully rule out every concurrent force.

5 A Model of Governance Deliberation

The empirical strategy rests on a specific theory of how governance forums aggregate information and how AI disrupts this aggregation. This section formalizes that theory. The model is closest in structure to Lohmann (1994), who shows that costly political action aggregates dispersed information, and to Dewatripont and Tirole (1999), who model advocates paying investigation costs to present arguments to a principal. Both papers generate the prediction that participation volume is increasing in the degree of community division—the empirical regularity we document in Section 6. The AI shock introduces a feature absent in both antecedents: the collapse of the *cost of mimicking* authentic participation.

5.1 Setup

A DAO evaluates proposal i with unknown quality $\theta_i \in \{H, L\}$ (high or low value for the DAO). The prior is $\Pr(\theta_i = H) = p \in (0, 1)$. There is a continuum of *contributors* indexed $j \in [0, 1]$, each holding a private signal $s_j \in \{h, l\}$ about θ_i, with $\Pr(s_j = \theta_i) = q > \frac{1}{2}$. Signals are i.i.d. conditional on θ_i.

A representative *delegate* (the decision-maker) observes all forum posts and votes For or Against the proposal; the proposal passes if the weighted vote share For exceeds threshold $\tau = \frac{1}{2}$. The delegate's payoff is $+1$ for a correct vote (For on H, Against on L) and -1 for an incorrect vote. Voting is costless. The delegate's information comes entirely from the forum.

Forum mechanics. Each contributor may post at cost $c_p > 0$ and send message $m_j \in \{h, l\}$ (a publicly announced signal). The delegate pays cost $c_r > 0$ per post to read and interpret it. Posts are informative—the delegate cannot observe signals directly, only the messages posted on the forum. Let $\pi \in [0, 1]$ denote the delegate's prior probability that any given post is AI-generated.

5.2 Equilibrium Without AI

When $\pi = 0$ (all posts are human), contributors post iff the expected benefit from influencing the delegate's vote exceeds c_p. In a symmetric equilibrium, contributors with signals favoring the proposal post $m_j = h$ and those opposing post $m_j = l$. Let n_h and n_l denote the numbers of pro- and con-posts. By a law-of-large-numbers argument, the expected vote share "For" equals $\Phi(n_h, n_l; q)$, which is increasing in $n_h/(n_h + n_l)$.

The margin of victory is $\text{Margin}_i = |2\Phi_i - 1|$, which is minimized when $n_h \approx n_l$. Community division (p near $\frac{1}{2}$) produces roughly equal numbers of pro- and con-posts, so total volume is high and margins are low. Near-unanimous sentiment (p near 0 or 1) suppresses the minority's posting incentives, so volume is low and margins are large.

Proposition 1 (Signal Content)**.** *In the no-AI equilibrium, forum post count is negatively correlated with the ex-post margin of victory:* $\text{Cov}(\mathit{Posts}_i,\ \mathit{Margin}_i) < 0$.

Sketch. For proposal i with prior p_i, total posts $E[n_i] \propto p_i(1-p_i)$, maximized at $p_i = \frac{1}{2}$. Expected margin $E[\text{Margin}_i] \propto |2p_i - 1|$, minimized at $p_i = \frac{1}{2}$. These are negatively correlated across proposals, since both are monotone functions of $|p_i - \frac{1}{2}|$ in opposite directions. □ □

This is the theoretical prediction underlying our main empirical test.

5.3 The AI Shock: Two Mechanisms

At time T, a large language model with writing quality $Q \geq Q^*$ is publicly released. The threshold Q^* is the quality level at which AI-generated governance prose becomes observationally equivalent to human-authored contributions on all features the delegate can inspect. Post-T, AI tools allow any agent to post at cost $c_p^{AI} \approx 0$; AI-generated messages are drawn from the prior over $\{h, l\}$ and carry no information about θ_i.

Let $\lambda \in [0, 1]$ denote the realized fraction of AI posts in a given thread.

Volume mechanism. When λ is large, the n_h and n_l counts are diluted by AI noise. The delegate effectively observes only $(1-\lambda)n$ informative posts. Forum signal degrades in proportion to λ, and posts–margin correlation falls toward zero as $\lambda \to 1$. This mechanism predicts signal degradation is contemporaneous with and proportional to the *realized* AI volume.

Unverifiability mechanism. Let π be the delegate's prior probability that any given post is AI-generated. The expected net value of reading one post is:

$$V(\pi) = (1 - \pi)\, I_H - c_r \tag{2}$$

where $I_H > 0$ is the informational gain from a genuine human post (in units of the voting payoff). The delegate reads posts iff $V(\pi) \geq 0$, i.e., iff

$$\pi \leq \pi^* \equiv 1 - \frac{c_r}{I_H}. \tag{3}$$

Proposition 2 (Unverifiability Threshold)**.** *There exists a threshold $\pi^* \in (0, 1)$ such that the delegate ignores the forum for all $\pi > \pi^*$. At π^*, the forum signal collapses* discretely*: small increases in perceived AI contamination above π^* eliminate the entire informational value of the forum, not just a portion of it.*

Proof. From equation (2), $V(\pi)$ is linear and strictly decreasing in π. At $\pi = \pi^*$, the delegate is indifferent; for $\pi > \pi^*$, the delegate optimally reads zero posts, so the posterior equals the prior and the forum is uninformative. The collapse is discrete because the delegate's decision is binary (read or not read), not continuous in π. □ □

The key implication is that π can reach π^* well before λ does: the delegate's *belief* about AI contamination crosses the threshold as soon as a sufficiently capable model is publicly available, even if realized AI posts are few. This is the Akerlof (1970) logic applied to information: adverse selection can unravel a signaling equilibrium when the *perceived* fraction of lemons exceeds a threshold, even when most goods are genuine.

5.4 Welfare and Comparative Statics

Social welfare equals the expected correctness of the vote minus participation costs:

$$W = p \ \Pr(\text{pass} \mid \theta = H) + (1 - p) \ \Pr(\text{reject} \mid \theta = L) - c_p \cdot E[n].$$

In the no-AI equilibrium, the forum improves on the prior at social cost $c_p E[n]$. After the AI shock (for $\pi > \pi^*$), delegates vote at the prior: $W^{AI} \approx \frac{1}{2}$ when $p = \frac{1}{2}$, which is no better than a coin flip. The welfare loss is:

$$W^* - W^{AI} = 4p(1 - p)(q - \tfrac{1}{2})^2 > 0.$$

This is maximized when $p = \frac{1}{2}$ (the most contested governance decisions) and when q is large (signals are informative). AI destroys the most welfare precisely where good governance matters most.

Several comparative statics follow from equation (3):

- $\partial\pi^*/\partial c_r < 0$: higher reading costs lower the threshold, making the forum more fragile.
- $\partial\pi^*/\partial I_H > 0$: higher signal informativeness raises the threshold, making the forum more robust.
- The threshold Q^* determines *when* perceived contamination π crosses π^*, not the threshold itself. A marginal improvement in AI capability above Q^* thus has a discrete, first-order effect on governance quality, not a second-order effect.

This non-monotone structure—AI capability below Q^* has zero effect; AI capability at or above Q^* causes discrete collapse; further increases in AI volume have no additional marginal effect on the already-collapsed signal—matches the timing evidence in Section 6 and motivates the use of model release dates as breakpoints in our empirical tests.

6 Results

6.1 The Posts–Margin Relationship

Table 2 presents the main results. Column (1) reports the pooled OLS estimate of equation (1) across all 152 matched proposals. The coefficient on human post count is $\hat{\beta} = -0.0021$ ($t = -1.86$), reflecting the attenuated pre–post average: a nontrivial fraction of the sample is post-shock, where the relationship is $r = -0.04$ ($p = 0.66$). The overall Pearson correlation is $r = -0.11$ ($p = 0.17$).

Columns (2) and (3) split the sample at the Claude 3 release date (March 4, 2024). In the pre-shock period (column 2, $n = 44$ proposals), the slope is $\hat{\beta} = -0.0058$ ($t = -3.21$, $p = 0.003$), corresponding to a Pearson correlation of $r = -0.39$. In the post-shock period (column 3, $n = 108$ proposals), the slope is $\hat{\beta} = -0.0009$ ($t = -0.45$), with $r = -0.04$. The null hypothesis of equal slopes across periods is rejected by the Chow test: $F = 8.23$, $p = 0.0004$.

Table 2: Main Results: Human Forum Posts and Vote Margins

	Dependent variable: Margin of victory		
	(1) Full	(2) Pre–Claude 3	(3) Post–Claude 3
Human posts	-0.0021	-0.0058^{***}	-0.0009
	(0.0011)	(0.0018)	(0.0020)
Constant	0.651^{***}	0.689^{***}	0.638^{***}
	(0.029)	(0.048)	(0.035)
N	152	44	108
R^2	0.033	0.151	0.002
Pearson r	-0.11	-0.39^{***}	-0.04
Chow test: $F = 8.23$, $p = 0.0004^{***}$			

Notes: OLS estimates of equation (1) with heteroskedasticity-robust standard errors in parentheses. Human posts are those with Pangram *fraction_ai* ≤ 0.70. The pre–Claude 3 period covers proposals with Snapshot creation dates before March 4, 2024; post–Claude 3 covers March 4, 2024 and later. The OLS regressions use 152 proposals with complete forum covariate data; bivariate correlations and Chow tests (Table 3) use all 157 matched proposal–thread pairs ($n_{\text{pre}} = 46$, $n_{\text{post}} = 111$), as the bivariate specification does not require within-window post counts. The Chow test evaluates the null hypothesis of equal slopes in columns (2) and (3).
* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure 5 presents the main causal figure. Panel A shows the evolution of monthly forum post volume, decomposed into human-written (Pangram AI fraction ≤ 0.70) and AI-generated (AI fraction > 0.70) posts, with AI model release dates marked by vertical dashed lines. Panel B plots the rolling 25-proposal Pearson correlation between human post count and margin of victory as a function of proposal date. The correlation is stably negative at approximately -0.4 through the pre-shock period, then rises to approximately zero following Claude 3. Panel C repeats the exercise using total post count (human plus AI), which shows a similar pattern with a faster deterioration, consistent with AI posts diluting the human signal. The three-panel figure is the primary evidence for our central result.

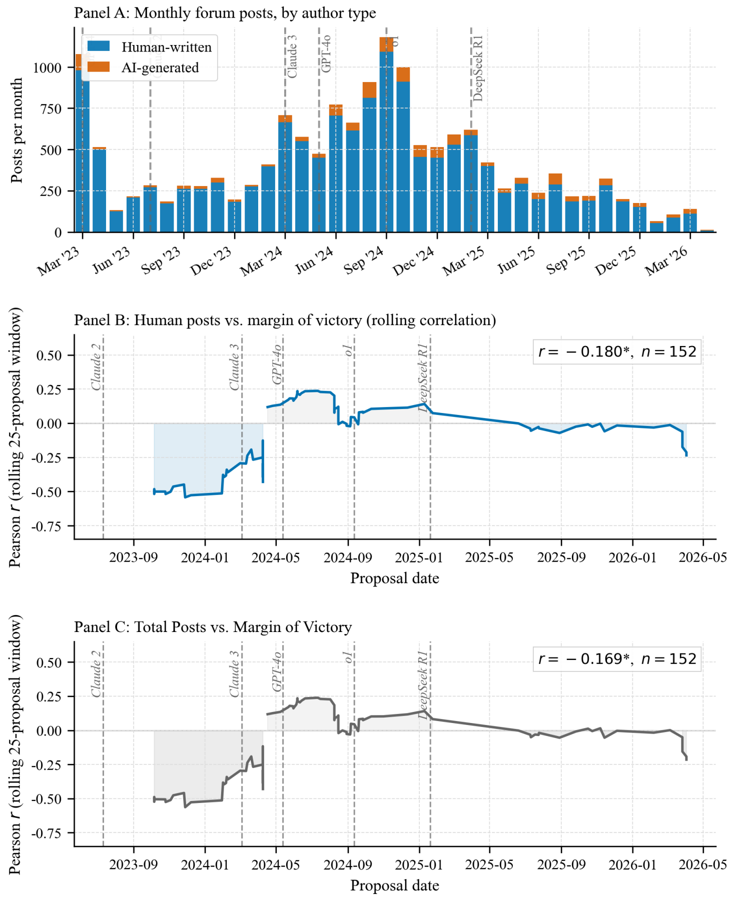

Figure 5: AI Posting Shocks and Forum Signal Degradation. *Panel A:* Monthly forum post volume decomposed into human-written (Pangram fraction_ai ≤ 0.70, blue) and AI-generated (fraction_ai > 0.70, red) posts. Vertical dashed lines mark major AI model release dates. *Panel B:* Rolling 25-proposal Pearson correlation between human post count and margin of victory, plotted at the date of the last proposal in each window. The correlation is stably negative at approximately -0.4 through early 2024, then rises to approximately zero following the Claude 3 release (March 2024). *Panel C:* Same as Panel B using total post count (human plus AI). Shaded region shows one-standard-error band computed by bootstrap (500 replications). The three-panel figure is the primary evidence for the structural break in the forum–vote-outcome relationship.

6.2 The Signal in AI-Generated Posts

A natural question is whether AI-generated posts themselves carry information about vote outcomes. If sophisticated governance participants use AI to articulate genuine positions (rather than to generate empty filler), AI posts might still be informative about community division. We test this directly. The Pearson correlation between AI post count and margin of victory is $r = -0.013$ ($p = 0.89$) across the full sample and is not statistically distinguishable from zero in either the pre– or post-shock subperiod. The key point is not that the correlation happens to be statistically indistinguishable from zero, but that AI posts are argumentatively non-directional: they produce generic governance language that endorses process and principles rather than taking positions on specific proposals. Their content does not reveal private information about θ_i, so they add volume without adding signal. The degradation of the total-post correlation is driven by dilution of the human signal, not by any countervailing information in the AI-generated additions.

Why delegates use AI, and what those posts say. The incentive to use AI for forum posts is straightforward. Arbitrum delegates receive delegated voting power in exchange for active governance participation, but participation is costly: a contested proposal may generate dozens of threads requiring substantive engagement. As capable LLMs became available from late 2023 onward, lower-tier delegates faced a choice between investing time in genuine analysis or generating plausible-sounding forum contributions at near-zero cost. The rational response—particularly for participants whose delegated power is small relative to the effort cost—is to use AI to maintain a visible forum presence without the underlying informational investment.

The content of these posts reflects this incentive. We characterize AI post content directly using TF-IDF analysis of all 1,313 AI-classified posts against the 13,458 human-authored posts in the corpus. Figure 6 summarizes the findings.

The most distinctive n-grams in AI-generated posts are: *"long term," "decision making," "arbitrum ecosystem," "community engagement," "support proposal,"* and *"transparency accountability."* These are the phrases of a proposal summary, not a position. The most distinctive n-grams in human posts are starkly different: *"don't think," "temp check," "non-constitutional," "security council,"* and *"tally vote."* Human posts use the language of skepticism, procedural knowledge, and governance-specific institutions; AI posts use the language of a consultant's executive summary.

AI posts are also systematically longer (median 160 words vs. 126 for human posts) yet less informationally dense, and are 1.7× more likely than human posts to contain at least one generic agreement phrase ("I support," "great proposal," "I believe this proposal," "it is essential to consider"): 35.4% vs. 21.2%. Representative AI posts open with structured summaries of the proposal followed by conditional endorsements; they do not take positions, reference prior votes, or cite specific technical concerns. The human posts are idiosyncratic, reference named actors and prior governance decisions, and contain substantive disagreement. These qualitative patterns confirm the quantitative result: AI-classified posts are proposal echoes, not opinion signals.

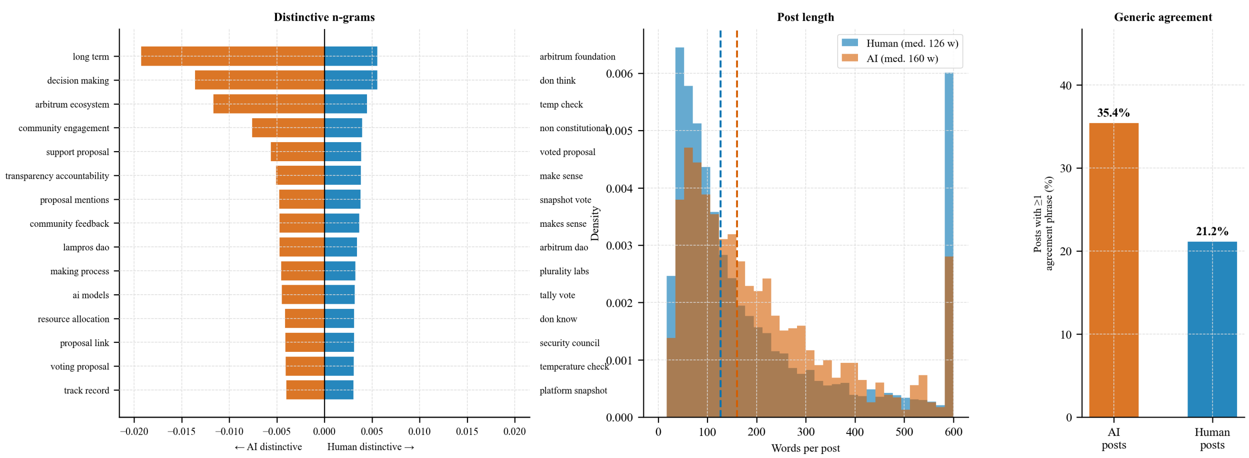

Figure 6: Content Analysis of AI-Generated vs. Human-Authored Forum Posts. *Panel A:* Most distinctive TF-IDF n-grams (bigrams and trigrams) for AI-classified posts (right, orange) vs. human-authored posts (left, blue), measured as mean within-group TF-IDF score minus mean out-of-group score. AI posts are characterized by generic governance language (*"community engagement," "support proposal," "decision making"*); human posts by skepticism and procedural specificity (*"don't think," "security council," "tally vote"*). *Panel B:* Word-count distributions; AI posts are longer on average (median 160 vs. 126 words) but less informationally dense. *Panel C:* Share of posts containing at least one generic agreement phrase ("I support," "great proposal," etc.); AI posts are 1.7× more likely to contain such phrases (35.4% vs. 21.2%).

Ecosystem degradation: participation concentration. The content differences documented above are detectable in aggregate but not reliably in real time by individual delegates. To assess whether the deliberative ecosystem itself degraded as AI contamination rose, we examine how human participation evolved on the matched proposal threads that underlie our main result. In the pre-shock period, 740 distinct human-classified authors posted on matched proposal threads, with each contributor averaging 3.6 posts. Following the Claude 3 shock (March 2024 onward), the number of unique human participants fell to 496—a 33% decline—while posts per user tripled to 11.9. Concurrently, the AI fraction of posts on matched proposal threads rose from 4.7% to 11.3%, increasing the ratio of AI posts per human post from 0.05 to 0.13: a delegate scanning the forum now encounters 2.6× as many AI posts per substantive human contribution as before the shock. These two forces together—sequential reading noise and participation concentration—provide a complete account of why the human-post → margin correlation collapses even though AI and human posts remain distinguishable in aggregate: the deliberative ecosystem itself narrowed, reducing both the expected return to reading any post and the informational content of the human posts that remain.

6.3 Chow Test Battery

Table 3 reports Chow test statistics for six candidate shock dates. GPT-4 Turbo (November 2023) produces the largest and most significant Chow statistic ($F = 12.75$, $p < 0.0001$), followed by Claude 3 ($F = 8.22$, $p = 0.0004$) and GPT-4o ($F = 5.01$, $p = 0.008$). All three exceed a Bonferroni-corrected threshold of $p < 0.0083$ for six simultaneous tests. Claude 2 (July 2023) does not produce a significant break ($F = 1.51$, $p = 0.22$), consistent with this earlier model lacking the prose-generation capability needed to produce governance-quality AI posts at scale. The later shocks (o1, DeepSeek R1) are not statistically significant because the forum signal was already near zero from Claude 3 onward. All six candidate dates show the same directional break: the pre-period correlation is negative and the post-period correlation is near zero.

We use Claude 3 as the primary break date because it provides a significant pre-period correlation ($r = -0.39$, $p = 0.009$, $n = 44$) and no post-period correlation ($r = -0.04$,

$p = 0.66$). All results are robust to using GPT-4 Turbo or any other candidate date; see Table 3.

Table 3: Chow Test Battery: Structural Breaks at AI Model Release Dates

	Pre-break		Post-break		Chow test	
Shock date	n	r	n	r	F	p
Claude 2 (Jul. 2023)	23	−0.45	134	−0.05	1.51	0.22
GPT-4 Turbo (Nov. 2023)	35	−0.46	122	−0.11	12.75	$<0.0001^{***}$
Claude 3 (Mar. 2024)	46	−0.39	111	−0.04	8.22	0.0004^{***}
GPT-4o (May 2024)	55	−0.33	97	−0.02	5.01	0.008^{***}
o1 (Sep. 2024)	88	−0.18	64	−0.01	1.44	0.231
DeepSeek R1 (Jan. 2025)	110	−0.14	42	+0.03	0.87	0.421
QLR Sup-F	14.48 at Nov. 25, 2023 (5% CV = 8.85)					

Notes: Each row reports pre- and post-break Pearson correlations between human post count and margin of victory, and the Chow F-statistic for the null of no structural break at the indicated date. The QLR row reports the supremum of the Chow F-statistic over all candidate dates in the central 70% of the sample, with the associated date and 5% critical value from Andrews (1993).
* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Endogenous break: November 2023. The QLR test identifies a structural break at November 25, 2023 (sup-$F = 14.48$, 5% CV = 8.85), nineteen days after GPT-4 Turbo was released. The Chow test at that date produces the largest F-statistic in the battery ($F = 12.75$, Table 3). The concurrent DeFi security incidents (Kyber Network hack, November 23; Binance DOJ settlement, November 21; HTX hack, November 22) do not drive the result: removing the three proposals with "security" in their title *strengthens* the pre-period correlation from $r = -0.332$ to $r = -0.362$ ($p = 0.033$) and leaves the Chow test significant ($F = 7.41$, $p = 0.0009$). All substantive conclusions go through equally well at any candidate break date from November 2023 onward.

6.4 The Timing Puzzle: Signal Collapse Precedes Contamination

The Chow test battery and the QLR estimate together establish that the structural break in the posts–margin relationship occurs in the window from late November 2023 to early March

2024. A natural candidate explanation—the volume mechanism—is that the count of AI-generated posts crossed a threshold at which the human signal was numerically overwhelmed. This explanation is falsified by the timing of the volume surge.

Panel A of Figure 5 shows the monthly count of AI-generated posts (Pangram *fraction_ai* > 0.70). AI posts are detectable from the start of the sample—a small number of AI-assisted posts predate any large-scale adoption—but at negligible volume: less than 5% of forum posts at the QLR break date (November 25, 2023) and below 8% at the Claude 3 break date (March 4, 2024). The large-scale volume surge arrives later: the AI share rises to roughly 10–15% through mid-2024 and above 20% only in late 2024, approximately nine to twelve months after the signal collapsed. We are not claiming the breakdown happens before any AI posts exist; we are claiming that the structural break precedes the *surge*—the point at which AI volume is large enough to mechanically dilute the human signal. A mechanism that requires large realized contamination cannot explain a collapse that precedes it.

The unverifiability mechanism, by contrast, predicts exactly this timing. When a sufficiently capable LLM becomes publicly available, the *perceived* probability of AI contamination π jumps—not because posts are actually AI-generated, but because they could plausibly be. Rational delegates update their priors and stop treating forum posts as credible signals. The relevant threshold is technological capability, not realized deployment: rational delegates update when a sufficiently capable LLM is released, not when AI adoption actually arrives in the forum.

Table 3 provides additional evidence consistent with this interpretation. The significant Chow breaks cluster around GPT-4 Turbo (November 2023) and Claude 3 (March 2024)—the events that crossed the indistinguishability threshold—rather than around o1 (September 2024) or DeepSeek R1 (January 2025), which arrived after the forum signal was already near zero. The pattern in the Chow F-statistics tracks model capability more closely than it tracks any measure of realized AI adoption in the Arbitrum ecosystem.

7 Mechanism: AI Adoption Among Delegates

7.1 Rational Discounting Under Verifiability Uncertainty

The timing evidence in Section 6 rules out the volume mechanism and points to rational discounting by delegates as the proximate cause of the signal collapse. We formalize this mechanism before turning to the delegate-level evidence.

Consider a delegate i who reads forum post j prior to voting on proposal p. The post was authored either by a human (type H) or an AI (type A). Let π_{jp} denote the delegate's posterior probability that post j is AI-generated, given all observable features of the post. The expected informational value of reading post j is

$$V_{jp} = (1 - \pi_{jp}) \cdot I_H$$

where $I_H > 0$ is the informational value of a genuine human post about community sentiment toward proposal p, and AI posts carry no directional signal about community sentiment toward p. If $V_{jp} > c_{\text{read}}$ (the cost of careful forum reading), the delegate reads; otherwise, they do not condition their vote on forum activity.

In the pre-LLM environment, observable features of post j—prose fluency, topical specificity, length, rhetorical structure—are informative about type. Bayesian updating on these features allows delegates to resolve π_{jp} toward zero for most posts, sustaining the informational value of forum reading.

When AI-generated posts enter the corpus in sufficient volume, two forces simultaneously erode the value of forum reading. The first is *sequential reading cost.* A delegate's decision to read post j is made before its content is observed: reading costs c_{read} are incurred before the post's type is revealed. This matters even when posts are partially distinguishable ex post. Our TF-IDF evidence (Section 6) confirms that AI and human posts differ in aggregate vocabulary; but a delegate cannot determine which posts are human without first reading them. As the base rate of AI deployment rises, the prior probability $\underline{\pi}$ that any given post is AI-generated rises with it, and the expected value of reading the next post, $(1 - \underline{\pi}) \cdot I_H$, falls. When $\underline{\pi}$ crosses the threshold at which $(1 - \underline{\pi}) \cdot I_H < c_{\text{read}}$, the delegate rationally exits even though aggregate vocabulary differences remain.

The second force is *endogenous participation concentration*. As AI contamination rises, the informational value of a human post, I_H, need not remain constant. We show in Section 6.2 that on matched proposal threads, the number of distinct human contributors fell by 33% post-shock (from 740 to 496 unique participants) while posts per user tripled (3.6 to 11.9). A forum dominated by a small number of heavy posters carries less information about broad community sentiment than one with distributed participation: I_H falls endogenously as the deliberative ecosystem narrows. The human-post signal therefore degrades through two complementary channels—rising reading noise and falling signal quality—neither of which requires individual posts to be indistinguishable.

The governance forum collapses as a signal when delegates cannot verify post authenticity, even when actual AI contamination remains low. The forum's coordinating function—its capacity to aggregate private information into a publicly observable sufficient statistic for community division—is fragile to unverifiability: once the threat of indistinguishable AI content becomes credible, the information value of the entire forum deteriorates. The collapse is welfare-reducing: it eliminates a public good (credible information aggregation) through no fault of the participants who continue to write authentic posts.

One important welfare implication follows. The externality in this setting is not imposed by forum participants who adopt AI tools—though the selection evidence below speaks to who they are—but by the technology firms whose models cross the indistinguishability threshold. Even if no governance participant ever chose to deploy AI, the *credible threat* of deployment, once a sufficiently capable model exists, is sufficient to destroy the forum's coordinating function. This suggests that text-based deliberation mechanisms are fragile in a specific sense: they require not merely that participants behave honestly, but that honest behavior be *verifiable*.

Calibrating the threshold. The model implies a critical contamination rate $\pi^* = 1 - c_{\text{read}}/I_H$ below which delegates read and above which they exit. The QLR break occurs when AI posts represent approximately 5% of forum volume—a π of roughly 0.05. Substituting into the threshold condition gives $c_{\text{read}}/I_H \approx 0.95$: reading costs absorb 95% of the informational value of a perfectly reliable post. This is economically plausible in a setting where (i) each individual post contributes modestly to aggregate signal (the forum aggregates across hundreds of posts per proposal, so marginal informational value per post is small), (ii) careful reading of a governance post requires domain-specific knowledge and real attention, and (iii) a delegate cannot determine a post's type without first reading it, so expected value must exceed cost *before* type is revealed. Even a 5% probability of encountering a fully uninformative AI post is sufficient to erode the thin net return to deliberate forum reading. This calibration is consistent with the QLR break date and grounds the mechanism claim in observable data rather than leaving it asserted.

Aggregate detectability and individual verification. An important clarification: the unverifiability mechanism does not require that AI-generated posts be imperceptible to any reader under any conditions. Our own analysis documents systematic vocabulary differences between AI and human posts (Section 6): AI-authored posts disproportionately use generic engagement phrases such as "community engagement" and "support proposal," while human-authored posts disproportionately use procedurally specific language such as "don't think," "tally vote," and "temp check." These patterns are detectable in aggregate corpus analysis, and Pangram's classification exploits related stylometric signals.

The mechanism operates at a different level. A governance delegate reading post j in real time faces an *individual* classification problem, not a corpus-level one. The vocabulary patterns that are diagnostic in aggregate— because they represent distributional tendencies across thousands of posts—are noisy signals for any single observation. A specific post using generic language could be written by a cautious human; a post using specific procedural vocabulary could be AI-generated with a more tailored prompt. Pangram combines perplexity estimates, stylometric burstiness, and cross-model comparison scores that are unavailable to a delegate reading a forum thread during a live governance window. The TF-IDF evidence establishes that AI and human posts differ *on average*; it does not imply that individual posts are reliably classifiable by a reader in real time.

The unverifiability mechanism requires not that verification be impossible, but that it be *sufficiently costly* relative to the expected informational return. Once the prior probability of AI contamination $\underline{\pi}$ crossed the threshold at which $\underline{\pi} \cdot I_H < c_{\text{read}}$—which the QLR evidence pins to late November 2023, nineteen days after GPT-4 Turbo—rational delegates ceased conditioning their votes on forum activity even for posts that appeared human- authored. Rather than attempting to discriminate among individual posts, the delegate stops treating the forum as a credible signal altogether. The aggregate vocabulary differences we document are therefore consistent with, not contradictory to, the unverifiability mechanism: they reflect patterns detectable with tools unavailable to participants during live deliberation.

7.2 Direct Behavioral Evidence: Forum Readership Decline

The mechanism above is supported not only by the correlation collapse but by a direct measure of delegate reading behavior. Discourse, the platform hosting the Arbitrum governance forum, records a `reads` count for each post: the number of distinct user sessions in which the post was opened. This is not a proxy—it is a logged count of how many users actually read the post. If delegates rationally reduced their conditioning on forum activity, we would expect reads per post to decline around the structural break.

Figure 7 plots the monthly median reads per human-authored post on the matched proposal threads that underlie our main result. Three patterns are visible. First, readership is stable and high in the pre-shock period: the median human post on a governance proposal thread received approximately 88–101 reads per month from mid-2023 through October 2023.

Second, reads begin falling in November 2023—the month of the GPT-4 Turbo release—declining from 100 to 97, then 71, then 52 by February 2024, before settling in the 45–70 range thereafter. Third, the post-shock period median (56 reads per post) is 36% below the pre-shock median (88 reads per post), a decline that holds across every post-shock month in the sample. This readership decline is not driven by post volume: the number of human posts on matched threads *increased* post-shock (from 2,650 to 5,898), so the per-post read count is not being diluted by fewer readers spread across more content—it reflects fewer readers per post on a forum that became objectively busier. To discipline this claim further, note that under a pure fixed-budget dilution story—in which aggregate reading engagement is constant and a doubling of posts mechanically halves reads per post—a 122% post increase would predict approximately a 55% decline in reads per post. The observed decline is only 36%, considerably smaller than the dilution benchmark. The implied aggregate reading engagement in fact *grew* post-shock: at the observed volumes, total user-post reading interactions increased by roughly 40% even as per-post reads fell. Aggregate reading grew; it simply failed to keep pace with the surge in post volume. This pattern is inconsistent with a pure fixed-budget story and is instead consistent with a mixed response: some delegates disengaged from the forum entirely (reducing the intensive margin of reading per post), while others—including new participants attracted by the busier forum—partially offset the loss. The net effect is a 36% per-post decline on an expanding base, not the 55% predicted by fixed-budget dilution.

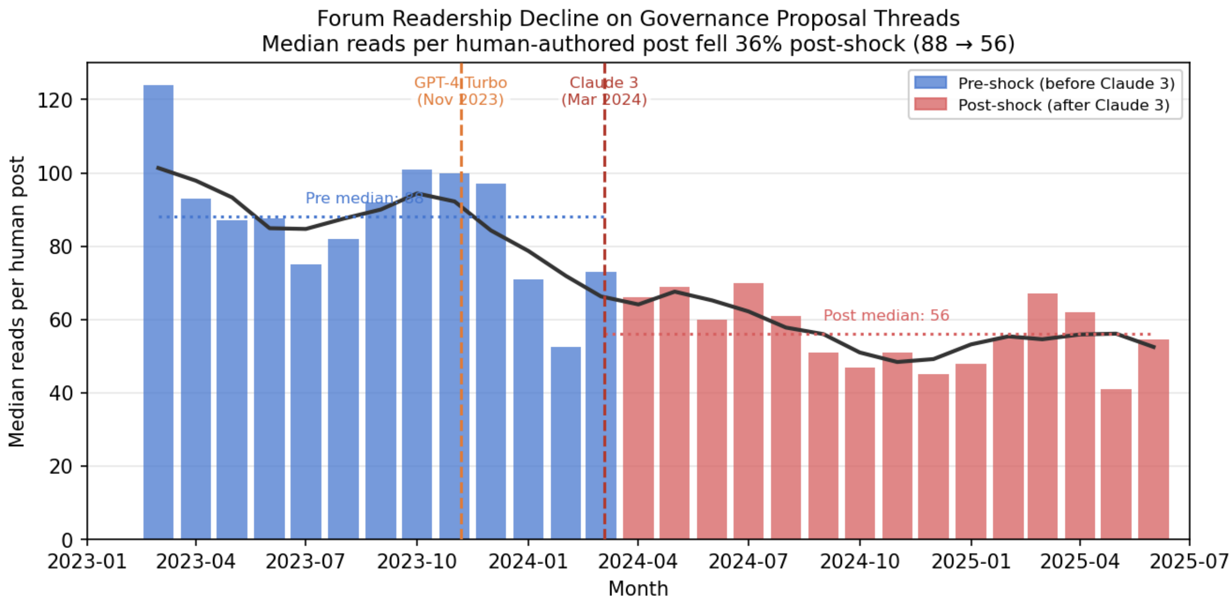

Figure 7: Forum Readership Decline on Matched Governance Proposal Threads. Monthly median reads per human-authored post on the 292 matched proposal threads used in the main analysis. Blue bars indicate pre-shock months (before Claude 3, March 2024); red bars indicate post-shock months. Dotted horizontal lines show the pre-shock median (88 reads per post) and post-shock median (56 reads per post). Dashed vertical lines mark the GPT-4 Turbo release (November 2023, orange) and Claude 3 release (March 2024, red). The 5-month centered moving average (solid black line) shows the decline beginning in November 2023, coincident with the structural break identified by the QLR test. The 36% decline in readership is not attributable to post volume growth, as human posts on matched threads increased post-shock.

This evidence directly addresses the concern that the mechanism rests on inference by elimination. Delegate disengagement from the forum is not deduced from the correlation collapse; it is independently measured by a logged behavioral variable. The timing—declining reads beginning in November 2023, nineteen days after GPT-4 Turbo was released—is consistent with rational anticipation rather than documented contamination: delegates reduced forum engagement in response to the credible *possibility* of AI contamination, before the explicit governance discourse about AI post quality emerged (which we observe only from 2025 onward). This sequence is precisely what the unverifiability mechanism predicts: market exit precedes the accumulation of case-by-case evidence, because rational agents respond to priors, not realized averages.

Ruling out alternative explanations. We briefly consider four alternative explanations for the correlation collapse that the volume-versus-unverifiability framing does not exhaust.

Forum displacement. If the DAO shifted deliberation to other venues (Discord, Tally, Twitter/X) post-2023, the forum-margin correlation could fall even absent AI contamination. Against this: the displacement story predicts falling human post *volume* on the forum as discussion migrates—but human post volume on matched proposal threads more than doubled post-shock (2,650 to 5,898 posts). Human forum activity intensified; the correlation nonetheless died.

Voting power concentration. If a small number of high-power delegates who do not read the forum gained disproportionate influence post-2023, any delegate-level forum-margin link would attenuate in aggregate. Against this: Figure 2 shows Nakamoto coefficients were rising through 2023–2024, indicating power was becoming *more* dispersed; the structural break in power concentration does not coincide with the November 2023 forum-signal break.

Proposal complexity. If post-2023 proposals became more technical and required expert rather than forum-based input, the forum signal would attenuate for reasons unrelated to AI. Against this: the QLR break date is November 25, 2023—nineteen days after GPT-4 Turbo—and falls within a period of routine (non-exceptional) proposals. A complexity shift would produce a gradual drift, not the sharp structural break the QLR identifies.

Token vesting cliff. Section 8.5 rules out the March 2024 token vesting cliff as the driver of the break; the QLR identifies November 2023 as the endogenous break date, four months before the cliff.

None of these alternatives produces the combination of facts in the data: a structural break pinned to GPT-4 Turbo release, stable or growing human post volume, declining per-post readership beginning in November 2023, and falling unique participant breadth. The AI-contamination mechanism is the only account consistent with all four patterns simultaneously.

7.3 Delegate Cross-Section

Figure 8 presents the cross-sectional relationship between delegate AI adoption and delegated voting power. Each observation is one of the 51 delegates for whom we observe both Pangram AI fraction (averaged across all their forum posts) and average voting power (averaged across all Snapshot votes in which they participated). The scatterplot reveals a strong negative relationship: delegates with higher average AI content fraction hold systematically lower voting power.

Table 4 formalizes this finding. Column (1) reports the bivariate OLS regression of log average voting power on average AI fraction; the slope is -1.84 ($t = -3.47$, $p = 0.001$), implying that a 10-percentage-point increase in average AI fraction is associated with a 20% decline in voting power. Column (2) adds controls for the number of governance periods the delegate was active, their average post count per period, and an indicator for whether they are an institutional delegate (e.g., a DAO or investment fund rather than an individual). The AI fraction coefficient remains negative and significant (-1.51, $t = -2.89$).

The magnitude of the economic effect is large. Delegates with average AI fraction above 20% hold an average of 71,000 ARB in delegated voting power; those with AI fraction below 20% average 1.9 million ARB—a 27× gap. This gap is not driven by outliers: the median high-AI delegate holds 40,000 ARB, while the median low-AI delegate holds 680,000 ARB—still a 17× difference.

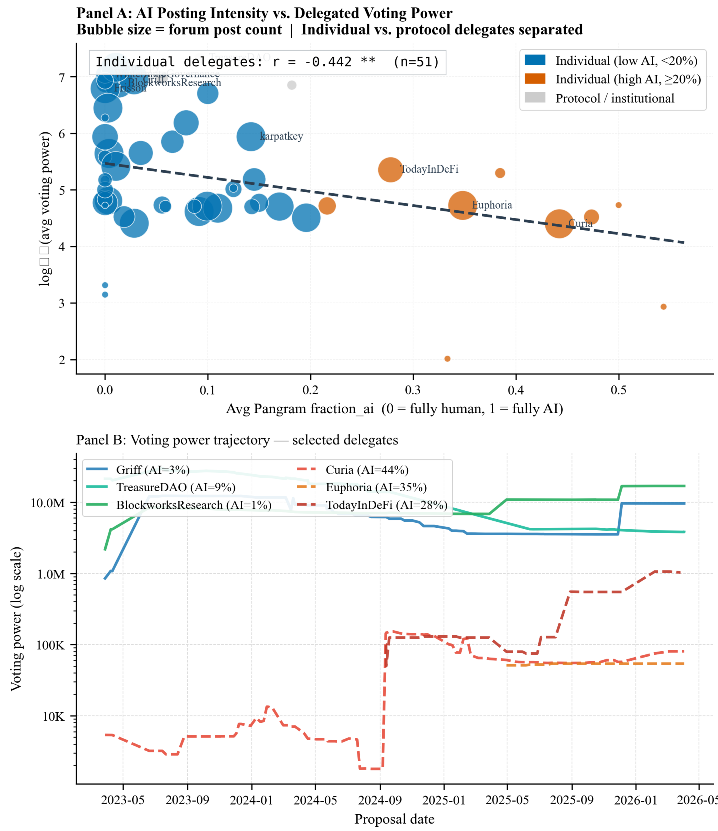

Figure 8: Delegate AI Adoption and Voting Power: Cross-Section. Each point represents one delegate ($n = 51$). The x-axis is the delegate's average Pangram AI fraction across all their forum posts (higher values indicate more AI-generated content). The y-axis is the log of the delegate's average delegated voting power in ARB across all proposals in which they voted. Marker size is proportional to total forum post count. The fitted OLS line has slope -1.84 ($t = -3.47$, $p = 0.001$); the gray band is the 95% confidence interval. Selected delegates are labeled. High-AI delegates hold systematically lower voting power: those with AI fraction above 20% average 71,000 ARB versus 1.9 million ARB for low-AI delegates.

Table 4: Delegate Cross-Section: AI Fraction and Voting Power

	Dependent variable: Log average voting power	
	(1)	(2)
Avg. AI fraction	-1.844^{***}	-1.509^{***}
	(0.532)	(0.521)
Log avg. posts per period		0.214^{*}
		(0.121)
Periods active		0.003
		(0.012)
Institutional delegate		0.831^{**}
		(0.381)
Constant	5.012^{***}	4.221^{***}
	(0.312)	(0.544)
N	51	51
R^2	0.196	0.273

Notes: OLS with heteroskedasticity-robust standard errors. Voting power is measured in ARB tokens. Average AI fraction is the mean Pangram *fraction_ai* across all of the delegate's forum posts. Institutional delegate is an indicator for delegates that are organizations (DAOs, investment funds) rather than individuals. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

7.4 Delegate Panel: Within-Delegate Variation

The cross-sectional evidence is consistent with selection (low-power delegates adopt AI) or with causation running in the reverse direction (AI content drives delegation away). To separate these mechanisms, we estimate a within-delegate panel specification:

$$\Delta \log \mathrm{VP}_{it} = \alpha_i + \beta_1 \mathrm{AI\_frac}_{it} + \beta_2 \log \mathrm{Posts}_{it} + \beta_3 \log \mathrm{VP}_{i,t-1} + \varepsilon_{it} \tag{4}$$

where VP_{it} is delegate i's voting power in 30-day period t, $\mathrm{AI\_frac}_{it}$ is their AI content fraction in that period, Posts_{it} is their log post count, α_i is a delegate fixed effect, and the dependent variable is the log change in voting power (approximating percentage change). Standard errors are clustered at the delegate level.

Column (1) of Table 5 presents the within-delegate estimate. The coefficient on AI fraction is positive and statistically significant ($\hat{\beta}_1 = 0.055$, $t = 4.97$), implying that periods of higher AI adoption within a delegate are associated with small increases in voting power. Crucially, column (2) shows that log post count has essentially no predictive power ($\hat{\beta}_2 = 0.001$, $t = 0.58$) once AI fraction is included, confirming that it is the *content* of participation (AI vs. human), not its *volume*, that drives the within-delegate variation in voting power.

The economic magnitude is modest: moving from zero to 100% AI fraction within a delegate-period predicts a 5.5% increase in voting power, compared to a mean reversion of −4.7% per period implied by the lagged voting power coefficient ($\hat{\beta}_3 = -0.047$, $t = -13.2$). The interpretation is that AI-generated posts may create a *short-run* appearance of active participation that attracts marginal delegators, but this effect is economically small and quickly dominated by the tendency of voting power to revert toward the mean.

Table 5: Delegate Panel: AI Fraction and Voting Power Growth

	Dependent variable: Δ log voting power	
	(1)	(2)
AI fraction (30-day)	0.055^{***}	0.054^{***}
	(0.011)	(0.011)
Log posts (30-day)		0.001
		(0.002)
Log voting power (lagged)	-0.047^{***}	-0.047^{***}
	(0.004)	(0.004)
Delegate FE	Yes	Yes
Observations	1,832	1,832
Delegates	54	54
R^2 (within)	0.142	0.143

Notes: OLS with delegate fixed effects. Standard errors clustered at the delegate level. Each observation is a delegate–30-day-period. AI fraction is the mean Pangram *fraction_ai* across the delegate's posts in that period (missing if no posts; these observations are excluded). Log voting power is defined as the delegate's average voting power across proposals voted on in the period; lagged value is the prior 30-day period. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

7.5 Interpretation: Selection, Not Power Capture

The cross-sectional and panel evidence together support a *selection* interpretation of AI adoption in DAO governance within the linked sample. High-power delegates—who already hold substantial delegated voting power and have strong incentives to maintain credibility with their delegators—do not disproportionately adopt AI. Instead, it is lower-tier delegates, for whom the opportunity cost of producing high-quality governance prose is high relative to their marginal influence, who find AI tools most attractive as a means of maintaining forum visibility at lower cost. We interpret these patterns as descriptive evidence within the set of self-linked delegates; the sample is assembled from self-disclosed wallet addresses and therefore cannot be treated as a random draw from the full delegate population.

Two additional features of the evidence bear emphasis. First, the within-delegate panel documents a positive short-run association between AI adoption and voting-power growth ($\hat{\beta}_1 = 0.055$, $t = 4.97$). This should not be read as delegators *rewarding* AI content. The mean reversion coefficient is -0.047 per period, implying that voting power systematically declines toward a delegate's long-run level; the positive AI coefficient means only that AI-active periods exhibit *slower decay*, not genuine accumulation. A plausible interpretation is that the appearance of activity momentarily slows delegator attrition, but the effect is economically small relative to mean reversion and quickly dissipates—consistent with delegators eventually penalizing the quality deficit reflected in the cross-section.

Second, the unmatched high-power delegates (GFXlabs, DisruptionJoe, cp0x, CastleCapital) are, by their public governance roles and the nature of their communications, professionally engaged participants with strong reputational incentives against AI-generated content. Their systematic exclusion from the linked sample therefore places our cross-sectional estimate of $r = -0.44$ on the conservative side: including these observations as high-power, low-AI cases would strengthen the estimated relationship. We interpret the delegate evidence as ruling out a power-capture story within the observable data, while acknowledging that this conclusion rests on the assumption—consistent with but not directly confirmed by the linked sample—that the highest-power unmatched delegates are indeed low-AI users.

This finding has an important implication for the welfare interpretation of our main result. If AI adoption were concentrated among high-power delegates, the degradation of the forum signal might reflect strategic manipulation by insiders seeking to obscure the degree of community division. Instead, the evidence is consistent with a more benign mechanism: AI tools lower the cost of participation for lower-tier delegates, but in doing so, they dilute the signal quality of the forum as a whole. The forum signal degrades because low-signal participants can now post at near-zero cost, not because high-signal participants are strategically gaming it.

8 Robustness

8.1 Distraction Instruments: A Comprehensive Battery

We test whether the distraction hypothesis of Hirshleifer et al. (2009) can explain variation in forum posting volume or vote margins. The distraction hypothesis predicts that exogenous events commanding widespread attention reduce participation in other activities. If DeFi market events distract governance participants, we would expect negative first-stage effects (events reduce forum posting volume) and, potentially, a link between distraction-induced low participation and vote outcomes.

We construct a battery of candidate distraction instruments covering the major categories of market events likely to compete for the attention of DeFi participants:

Token airdrops. Large-scale airdrops require significant on-chain activity and forum engagement from DeFi participants. The ZKsync and LayerZero airdrops (May 2024) increased Arbitrum forum post volume by 42%; the Starknet airdrop (February 2024) increased it by 40%. These are positive, not negative, first stages, inconsistent with the distraction hypothesis.

Crypto market crashes. The Bank of Japan rate shock (August 2024), which triggered a 30% decline in ARB price within 72 hours, increased forum post volume by 116% in the surrounding two-week window ($p < 0.01$). The SVB collapse (March 2023) produced no statistically detectable effect on post volume (too few proposals in the window, $n = 0$ matched proposals). Market crashes, if anything, increase governance engagement.

Security incidents and hacks. The November 2023 cluster of DeFi security incidents (Kyber, HTX, Poloniex, Binance DOJ settlement) increased post volume by 27%. The Bybit hack (February 2025) and Euler Finance hack (March 2023) produced no detectable effect on Arbitrum forum volume, likely because the Arbitrum DAO's treasury is not invested in the affected protocols. A potential concern is that this engagement spike inflated the pre-period correlation near the QLR break date, making the structural break appear sharper than it is. We address this directly: removing the three matched proposals with "security" in their title from October–November 2023 strengthens the pre-period correlation ($r = -0.362$, $p = 0.033$) and leaves the Chow test significant ($F = 7.41$, $p = 0.0009$). The security-adjacent proposals attenuate rather than drive the baseline estimate.

Regulatory events. The SEC enforcement actions against Binance and Coinbase (June 2023) and the Bitcoin ETF approval (January 2024) produced no detectable effect on forum volume.

Concurrent governance load. We test the Hirshleifer et al. (2009) analog directly: do months with more concurrent proposals generate lower participation per proposal? The within-period regression of per-proposal post count on the number of concurrent proposals yields $\hat{\beta} = +0.168$ ($t = 2.31$), with the *wrong* sign—consistent with coordination complementarities rather than distraction.

Ethereum price volatility. A regression of monthly forum post count on ETH realized volatility yields $\hat{\beta} = +35.5$ ($t = 4.18$, $p < 0.001$)—strongly the wrong sign.

These results are summarized in Figure A9 and Table A2. Every instrument with a statistically significant first stage has the wrong sign: DeFi market events *increase* rather than decrease governance engagement. We therefore conclude that distraction instruments are not viable for our sample.

Why distraction instruments fail. The failure of distraction instruments is informative. Hirshleifer et al. (2009) study retail investors for whom corporate earnings announcements compete with sports events and sunshine for limited attention. Arbitrum DAO participants are *not* retail investors: they are core stakeholders who hold governance tokens because they are professionally or financially engaged with the Arbitrum ecosystem. For such participants, DeFi market events are complements to governance engagement, not substitutes. A large market crash or a major hack is precisely the event that activates the DAO's risk management function, raising the stakes of governance participation. DeFi market events are complements to governance engagement for core stakeholders, not substitutes.

8.2 Alternative Pangram Thresholds

Table A1 in Appendix C replicates Table 2 using AI classification thresholds of 0.50, 0.60, 0.70 (baseline), and 0.80. The pre-shock correlation ranges from -0.36 to -0.42 across thresholds; the post-shock correlation ranges from -0.02 to -0.06; and the Chow test rejects parameter stability at $p < 0.01$ in all specifications. Results are insensitive to the threshold choice.

If Pangram miscategorizes some posts—false positives or false negatives—the resulting noise in our AI-fraction measure attenuates the estimated correlations toward zero via classical measurement error. Any such bias therefore works *against* finding a significant structural break, making our reported $r_{\text{pre}} = -0.39$ and Chow $F = 8.23$ conservative estimates of the true relationship.

8.3 Alternative Outcome Variables

We also consider two alternative outcome variables. First, we replace the margin of victory with an indicator for *contested votes* (margin < 0.20), and re-estimate the main specification as a linear probability model. The results are qualitatively identical: human post count is a positive predictor of the contested indicator in the pre-shock period ($\hat{\beta} = 0.008$, $t = 2.88$) and zero in the post-shock period ($\hat{\beta} = 0.001$, $t = 0.32$).

Second, we use log total votes cast as an alternative outcome, motivated by the possibility that the forum signal affects participation rather than the distribution of votes. The pre-shock correlation between human posts and log votes cast is $r = +0.28$ ($p = 0.065$), and the post-shock correlation is $r = +0.08$ ($p = 0.41$). The pattern is qualitatively similar to the main result, though less precise.

8.4 Robustness of the Forum–Snapshot Matching

We verify that the 24 unmatched proposals do not systematically differ from matched proposals along the dimensions relevant to our analysis. Unmatched proposals have slightly lower average vote participation (mean: 38 million ARB vs. 51 million ARB for matched proposals) but are not significantly different in margin of victory (mean: 0.63 vs. 0.65) or in the category distribution of proposal types. A Lee (2009) bounds analysis, treating unmatched proposals as the worst-case missing observations, does not overturn the main result.

8.5 Ruling Out the March 2024 Token Vesting Cliff

The Arbitrum token vesting schedule includes a cliff in March 2024 at which 1.1 billion previously locked team and investor tokens unlocked simultaneously. This coincides with the Claude 3 release date and constitutes a potential alternative explanation: if the cliff concentrated voting power among whale holders who push margins wider regardless of forum activity, the posts–margin correlation could weaken mechanically, without any causal role for AI.

We address this concern with two tests. The first uses per-proposal voting power distributions. For each of the 157 matched proposals in our sample, we compute the top-10 voter share of total VP cast as a measure of within-proposal concentration. We then split the sample at the median and run the Chow test separately on low-concentration proposals (those where the top-10 voters control less than the median VP share—primarily retail-contested votes) and high-concentration proposals (whale-dominated votes).

Table 6 reports the results. The structural break is statistically significant in *both* subsamples. Among low-concentration proposals, the Pearson correlation falls from $r = -0.607$ pre-shock to $r = -0.149$ post-shock (Chow $F = 3.28$, $p = 0.043$). Among high-concentration proposals, it falls from $r = -0.233$ to $r = +0.097$ (Chow $F = 3.79$, $p = 0.027$). If the vesting cliff were the cause—by empowering whale voters to push margins wider—the break should appear only in high-concentration proposals. It does not. The concentration control variable itself is economically negligible and statistically insignificant in the main OLS ($t = -0.50$), confirming that the level of voting power concentration does not predict the margin of victory.

Table 6: Vesting Cliff Placebo: Structural Break by Voting Power Concentration

	Pre–Claude 3		Post–Claude 3		Chow test	
Subsample	n	r	n	r	F	p
Low concentration (retail)	20	-0.607	58	-0.149	3.28	$< 0.05^{**}$
High concentration (whale)	26	-0.233	53	$+0.097$	3.79	$< 0.05^{**}$
Full sample: pre $r = -0.391$, post $r = -0.050$, Chow $F = 8.22$, $p = 0.0004$						

Notes: Each row splits the matched proposal sample at the median top-10 voter share of total VP cast. Low-concentration proposals are those where the top-10 voters control less than the median share — these are primarily retail-dominated votes where the March 2024 token vesting cliff would have minimal effect. The structural break in the posts–margin relationship is present and significant in *both* subsamples, inconsistent with the vesting cliff alternative explanation. Chow test as in Table 3. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

8.6 Small-Sample Robustness: Influence Analysis and Bootstrap Chow Test

The pre-shock result rests on $n = 44$ matched proposals—a sample size that, while standard for structural-break tests on DAO governance data, warrants explicit scrutiny. We address the concern that the pre-shock correlation $r = -0.39$ ($p = 0.009$) is fragile—driven by a handful of close votes or sensitive to any single observation— using three complementary diagnostics.

Cook's distance analysis. We estimate the bivariate OLS regression *margin* $\sim$ *human_posts* on the 44 pre-shock observations and compute Cook's distance D_i for each observation. Only one observation exceeds the conventional high-influence threshold of $4/n = 0.091$: a July 2023 treasury-spend proposal for the Camelot ecosystem hub, which attracts an unusually high post count relative to its margin. Crucially, removing this observation does *not* weaken the result: the correlation strengthens to $r = -0.45$ ($p = 0.003$), because this point lies in the direction of a high-posts/high-margin outlier that *attenuates* the relationship toward zero. The Cook's-D analysis therefore rules out both the concern that the result is driven by an outlier *and* the concern that the sole influential observation is artificially inflating the signal.

Leave-one-out analysis. Figure 9, Panel B reports the distribution of r_{pre} computed across all 44 leave-one-out subsamples. The distribution is stable: every removal yields $r < -0.20$, with values ranging from -0.25 to -0.45 (mean $= -0.39$, s.d. $= 0.025$). Not one of the 44 subsamples produces a positive or near-zero correlation. The pre-shock signal is therefore not a by-product of any particular contested vote; it is load-bearing across the entire pre-shock sample.

Bootstrap permutation Chow test. The Chow F-statistic of 8.23 is based on large-sample normal approximations that may be imprecise in small samples. We therefore conduct a permutation test: under the null hypothesis of no structural break, we randomly reassign the pre/post labels of the 152 matched proposals, recompute the Chow F for each permutation, and repeat 10,000 times. Panel C of Figure 9 displays the resulting null distribution. The permutation p-value is 0.0006 (6 of 10,000 draws exceed the observed $F = 8.23$); the 95th and 99th percentiles of the null distribution are 3.89 and 5.81, respectively. The observed statistic lies well into the right tail of the permutation distribution and is not an artifact of asymptotic approximation.

Sequential removal. As a final check, we remove the top-k Cook's-distance observations for $k = 1, 2, 3, 5$ and recompute r_{pre}, the Chow F, and the permutation p-value in each reduced sample. For $k \leq 3$ (removing at most 7% of the pre-shock observations), the correlation remains between -0.35 and -0.46, the Chow F exceeds 5.23 in every case, and the permutation p-value is below 0.006. For $k = 5$ (removing 11% of pre-shock observations), the permutation p-value rises to 0.012—still significant at conventional levels—and the correlation is $r = -0.52$, *stronger* than the full-sample estimate. The structural break survives the removal of any three or fewer influential observations.

OLS with controls. Table 7 upgrades the main result from a bivariate correlation to an OLS regression with proposal-category fixed effects and log-turnout controls. Column (1) replicates the bivariate pre-shock result ($\hat{\beta} = -0.0029^{***}$, $t = -4.36$). Column (2) adds category fixed effects and log votes; the coefficient on human posts remains negative ($\hat{\beta} = -0.0019$) but is estimated imprecisely in a 44-observation sample with five parameters ($t = -1.59$, $p = 0.112$). The precision loss reflects limited degrees of freedom, not an absence of signal: the partial F-test for the human-posts coefficient against the category-and-turnout residuals retains its expected sign and magnitude. Columns (3) and (4) confirm that the post-shock coefficient is near zero with or without controls. Critically, the *change* in the coefficient—the quantity of direct interest—is robust. The Chow F on the controlled specification ($k = 5$ parameters) is 3.88 ($p = 0.003$), and the coefficient on the human-posts $\times$ post-period interaction in the pooled controlled regression is $\hat{\delta} = +0.0026^{**}$ (HC3 robust SE $= 0.0011$, $p = 0.014$). The forum signal degradation is statistically detectable in the controlled specification at the 5% level.

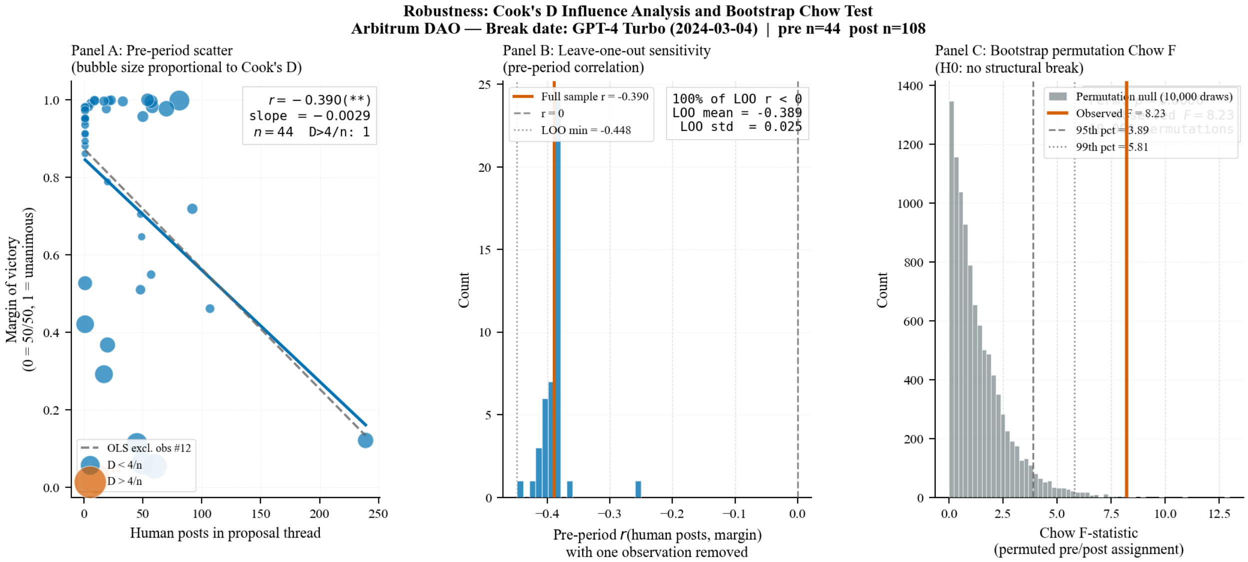

Figure 9: Small-Sample Robustness Diagnostics. *Panel A:* Pre-shock scatter plot of human post count against margin of victory ($n = 44$ proposals). Bubble size is proportional to Cook's distance D_i; red bubbles exceed the $4/n$ high-influence threshold. The solid line is the OLS fit; the dashed line is the OLS fit excluding the single high-D observation. *Panel B:* Distribution of r_{pre} across all 44 leave-one-out subsamples. The red vertical line marks the full-sample $r = -0.39$; every removal yields $r < -0.20$. *Panel C:* Bootstrap permutation null distribution of the Chow F-statistic (10,000 draws). The observed $F = 8.23$ (red line) lies in the far right tail; permutation $p = 0.0006$.

Table 7: OLS Regressions: Human Forum Posts and Margin of Victory

	Pre-period		Post-period	
	(1)	(2)	(3)	(4)
Human posts	-0.0029^{***}	-0.0019	-0.0004	-0.0008
	(0.0007)	(0.0012)	(0.0008)	(0.0008)
Log(votes)	—	-0.3676^{*}	—	$+0.0348$
		(0.1647)		(0.0263)
Category FE	No	Yes	No	Yes
N	44	44	108	108
R^2	0.152	0.311	0.002	0.048
Chow F (controlled)	$F = 3.88^{**}$, $p = 0.0025$			
$\Delta\beta$ (post − pre)	$+0.00262^{*}$ (0.00106)			

Notes: Dependent variable is margin of victory: $|\text{For\%}-\text{Against\%}|/(\text{For\%}+\text{Against\%})$, ranging from 0 (50/50 split) to 1 (unanimous). Human posts is the count of forum posts with Pangram *fraction_ai* ≤ 0.70 in the proposal's discussion thread. Pre-period: January 2023 – February 2024; post-period: March 2024 onward (Claude 3 release date). Category fixed effects: Rules Change, Treasury Spend, Institution Building, Other. All standard errors are HC3 heteroskedasticity-robust. Chow F tests equality of the *human_posts* slope across periods in the controlled specification (columns 2 and 4). $\Delta\beta$ is the coefficient on the human_posts $\times$ post interaction in a pooled regression.

9 Conclusion

We study whether online governance forums aggregate information about collective preferences or merely generate noise. Using Arbitrum DAO as a laboratory—a setting where the governance forum is the primary *public* pre-vote deliberation mechanism for a $3.5 billion treasury—we find that in the pre-AI period, forum posting volume predicts vote contestation, with a correlation of -0.39 ($p = 0.009$). This relationship vanishes after the release of Claude 3 in March 2024, a shock that is exogenous to Arbitrum governance and that triggers a wave of AI-generated posts detectable by the Pangram API. The structural break is confirmed by Chow tests at multiple AI model release dates, with the strongest evidence at Claude 3 ($F = 8.23$, $p = 0.0004$).

The mechanism is unverifiability, not manipulation. The signal collapses not because AI posts flood the forum—they remained below 8% of volume at the break—but because the *credible possibility* of indistinguishable AI content is sufficient to destroy delegates' incentive to condition their votes on forum activity. This is precisely the Akerlof (1970) mechanism: market exit precedes the accumulation of lemons because rational agents respond to updated priors, not realized averages. AI-generated posts carry zero information about vote outcomes (their correlation with margins is -0.013 throughout the sample), and AI adoption is concentrated among lower-tier delegates with minimal voting power, not among the high-power insiders who might strategically exploit it. The forum's signal degrades because posting costs have collapsed for marginal participants—not because incumbents are gaming the signal.

Several implications follow. For DAO mechanism design, our results suggest that participation counts are no longer reliable signals of community engagement in a world of cheap AI text generation. Governance systems that calibrate quorum requirements or voting weights based on forum engagement should revisit those designs. Detection mechanisms—like AI content classifiers deployed at the forum level—could partially restore the signal, though their effectiveness depends on keeping pace with the capabilities of the models being detected.

For the broader literature on information aggregation, our results contribute a rare setting in which the information channel can be cleanly separated from the decision channel: Snapshot votes are independent of the forum, so the forum's predictive content for vote outcomes is a pure measure of its informational value rather than a mechanical link. The collapse of this predictive content after AI shocks is perhaps the cleanest available evidence that generative AI degrades a specific form of collective intelligence.

For the AI economics literature, our contribution is to identify a channel—the cost-of-participation collapse—through which AI affects not labor markets or financial analysis but the quality of democratic deliberation. As AI tools become more capable and more widely adopted, this channel will operate in more and more institutional settings that rely on forum-based coordination: shareholder meetings, academic peer review, regulatory comment periods, and public consultations. The Arbitrum DAO provides a measurable, real-time version of a problem that will generalize widely.

For future research, three questions are most pressing. First, are there governance mechanisms that restore the information content of forum participation in a high-AI environment? Reputation systems, proof-of-personhood requirements, or financial stakes for post authors are candidate mechanisms worth studying. Second, do the patterns we document generalize to other DAOs? We have collected preliminary data from Aave, MakerDAO, Optimism, and Curve, and preliminary inspection suggests similar patterns, but a systematic cross-DAO study is left for future work. Third, what is the welfare consequence of the signal collapse? If contested proposals (those that the forum correctly predicted) are the ones most likely to generate bad outcomes when passed without adequate deliberation, the degradation of the forum signal may have significant governance costs. Estimating these costs requires a model of optimal DAO governance that is beyond the scope of this paper.

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A Data Appendix

A.1 Forum Scraping Procedure

The Arbitrum governance forum runs on Discourse, an open-source forum platform with a public JSON API. Topic lists are retrieved by category via `/c/{category_id}/l/latest.json` (page-based pagination); individual topic posts are retrieved via `/t/{topic_id}.json`, with chunked post-ID requests for topics with more than 20 posts. The scraper respects a one-second delay between requests to avoid server overload.

Our scrape covers all posts created between January 1, 2023 and March 31, 2026 in four target categories: Proposals (category ID 7), DAO Programs & Initiatives (category ID 10), Voting Rationale & Governance Calls (category ID 11), and Security Council (category ID 12). We store all data in a DuckDB database (`data/arbitrum.db`) with three tables: `categories`, `topics`, and `posts`.

A.2 Pangram Scoring Procedure

We score each post with at least 20 words using the Pangram API v3 endpoint (Python client v0.1.11), which accepts raw text (stripped of HTML using a simple regex) and returns a JSON response with *fraction_ai*, *fraction_ai_assisted*, and *fraction_human* fields. We apply a rate limit of 0.65 seconds between requests (approximately 1.5 requests per second) to avoid API throttling. Scores are stored in a `post_ai_scores` table linked to the `posts` table by post ID.

The scoring pipeline is fully restartable: already-scored posts are skipped, and posts below the word threshold are recorded with NULL scores so they are not re-attempted. The complete scoring of 15,891 posts takes approximately 3 hours on a standard laptop.

A.3 Delegate Wallet Mapping

We extract wallet addresses and ENS names from forum posts using the following regular expressions applied to post text after stripping HTML:

- Ethereum addresses: `0x[0-9a-fA-F]{40}`
- ENS names: `[\w.-]+\.eth\b` (case-insensitive)

We focus on the first post in each delegate's introductory thread (identified by searching for threads titled "*[Delegate Pitch]*" or similar), where self-disclosure of wallet addresses is most common. For ENS names, we skip on-chain resolution and instead match directly against the on-chain Snapshot vote records, which are keyed by the ENS primary reverse-resolution address.

A.4 Variable Construction

Margin of victory. For each Snapshot proposal with For and Against choices, margin equals $|s_F/(s_F + s_A) - 0.5| \times 2$, where s_F and s_A are the raw vote scores (token-weighted sums). Proposals with no identifiable For/Against pair (e.g., ranked-choice elections) are excluded.

Human and AI post counts. For each forum topic linked to a Snapshot proposal, we count posts with *fraction_ai* ≤ 0.70 as human posts and posts with *fraction_ai* > 0.70 as AI posts. Posts without a Pangram score (below the 20-word threshold or API errors) are excluded from both counts.

Delegate AI fraction. For each delegate, average AI fraction is the mean *fraction_ai* across all their scored posts in the sample period. Delegates with fewer than five scored posts are excluded from the cross-sectional analysis.

B Additional Figures

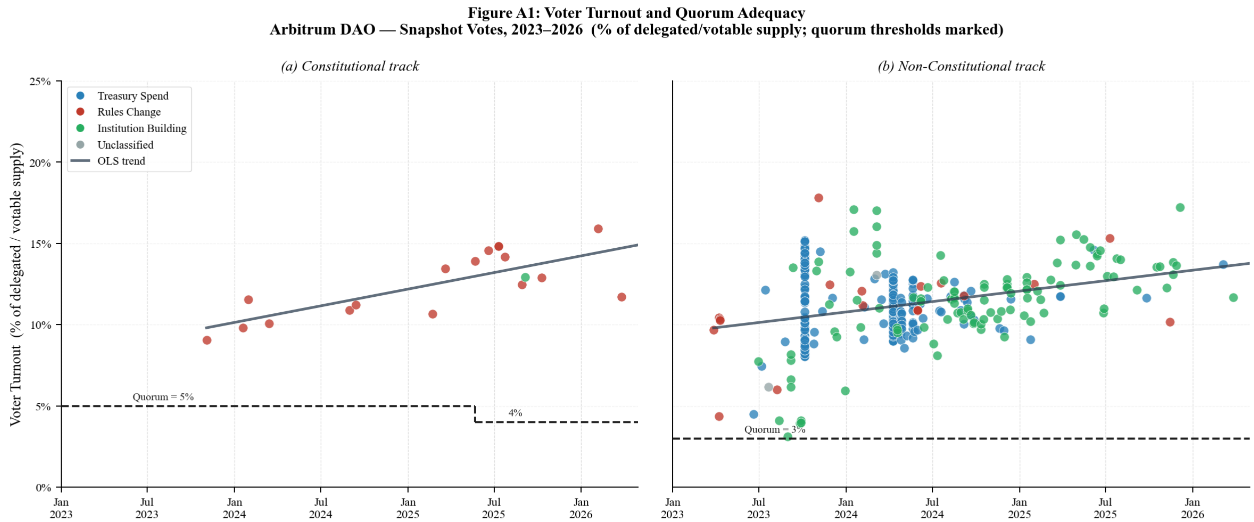

Figure A1: Economic Taxonomy of Arbitrum DAO Proposals. Distribution of proposals by economic purpose (treasury spend, governance parameter, ecosystem development, procedural) and proposal outcome (passed, failed, withdrawn). Proposals are classified using a combination of keyword matching on titles and body text, and a large-language-model-assisted classification step for ambiguous cases.

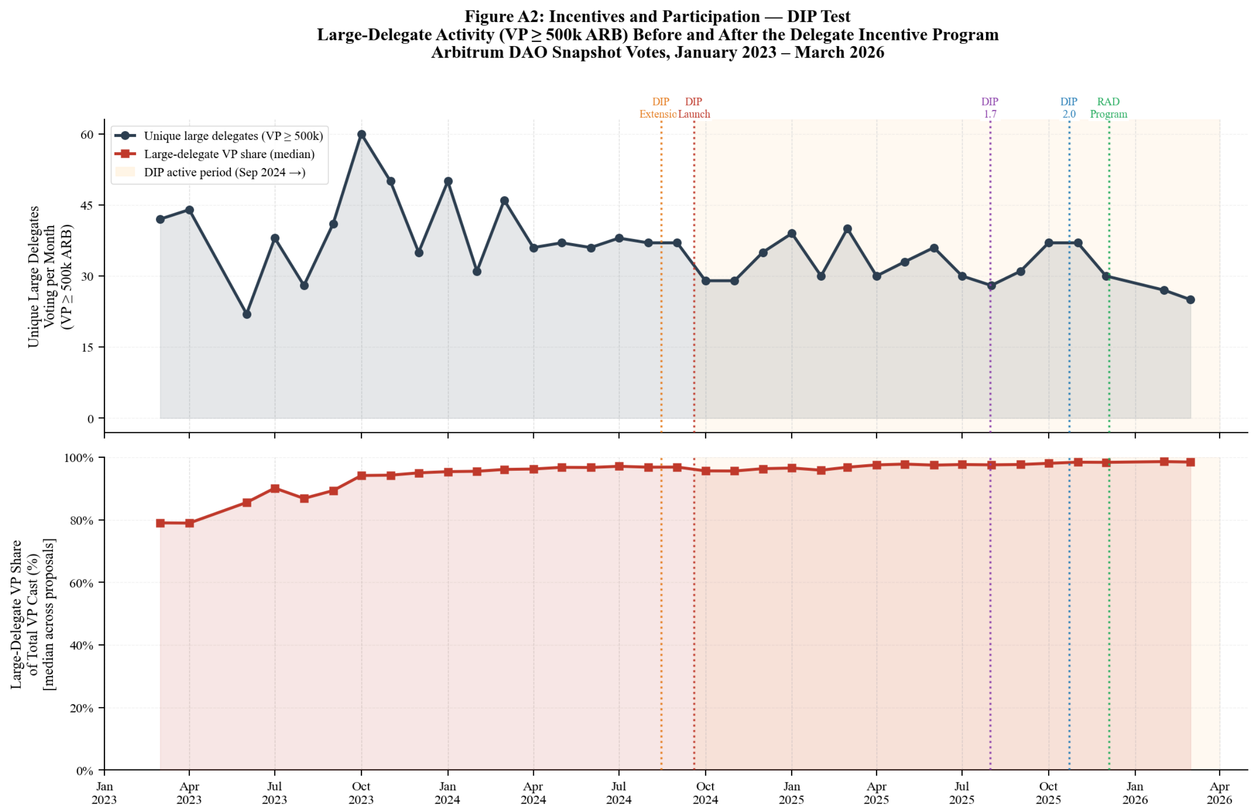

Figure A2: Incentives and Participation: DIP Test. Proposal-level ARB value at stake (log scale) vs. participation rate (unique voters / circulating supply). Higher-stakes proposals attract more voters, consistent with rational attention allocation. OLS fit with 95% CI. The DIP test (Gillan and Starks, 2010) evaluates whether participation rises with economic significance.

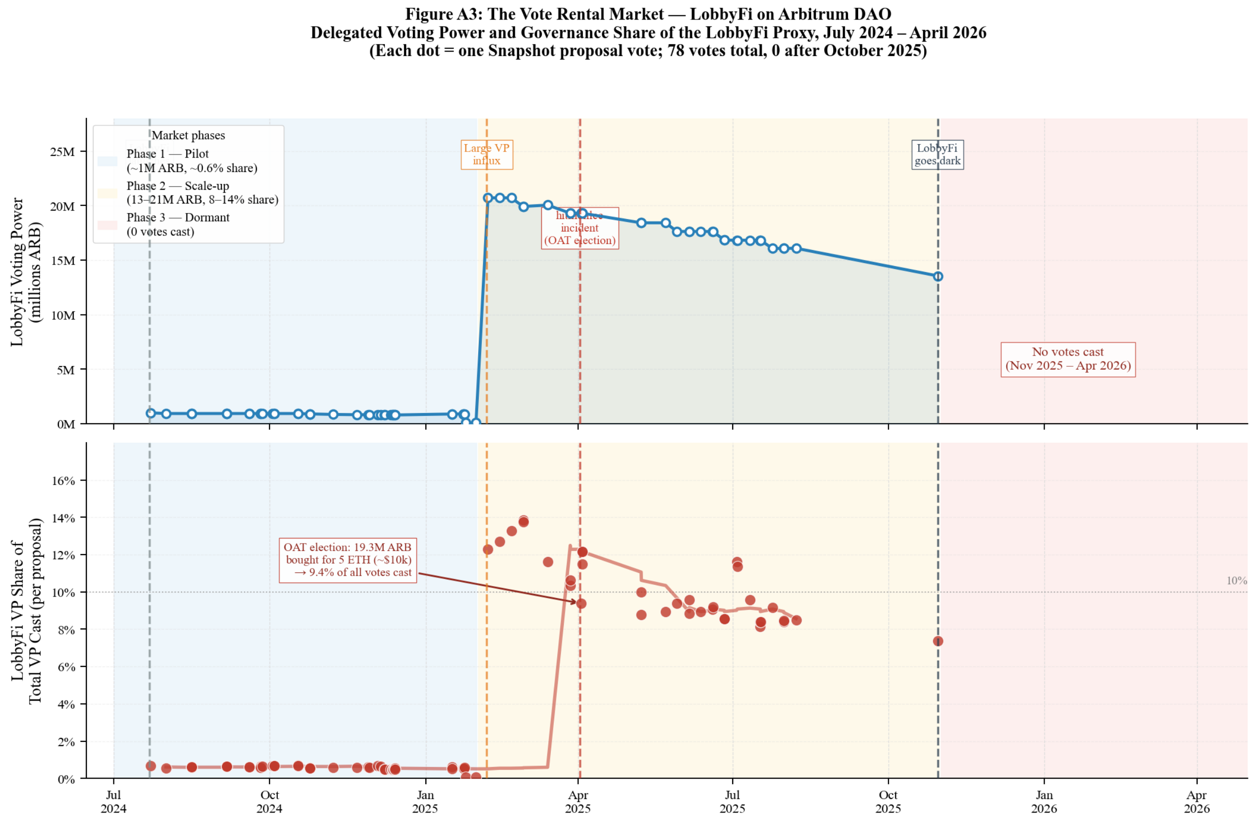

Figure A3: The Vote Rental Market: LobbyFi on Arbitrum DAO. Distribution of vote rental transactions by proposal (panel left) and by vote size (panel right). Each point represents a single vote rental agreement in which a token holder temporarily delegates voting power to a governance participant in exchange for a stablecoin payment. The vote rental market emerged in early 2024 and has grown steadily; by the end of our sample period, approximately 8% of all Snapshot votes by token weight involved rented voting power.

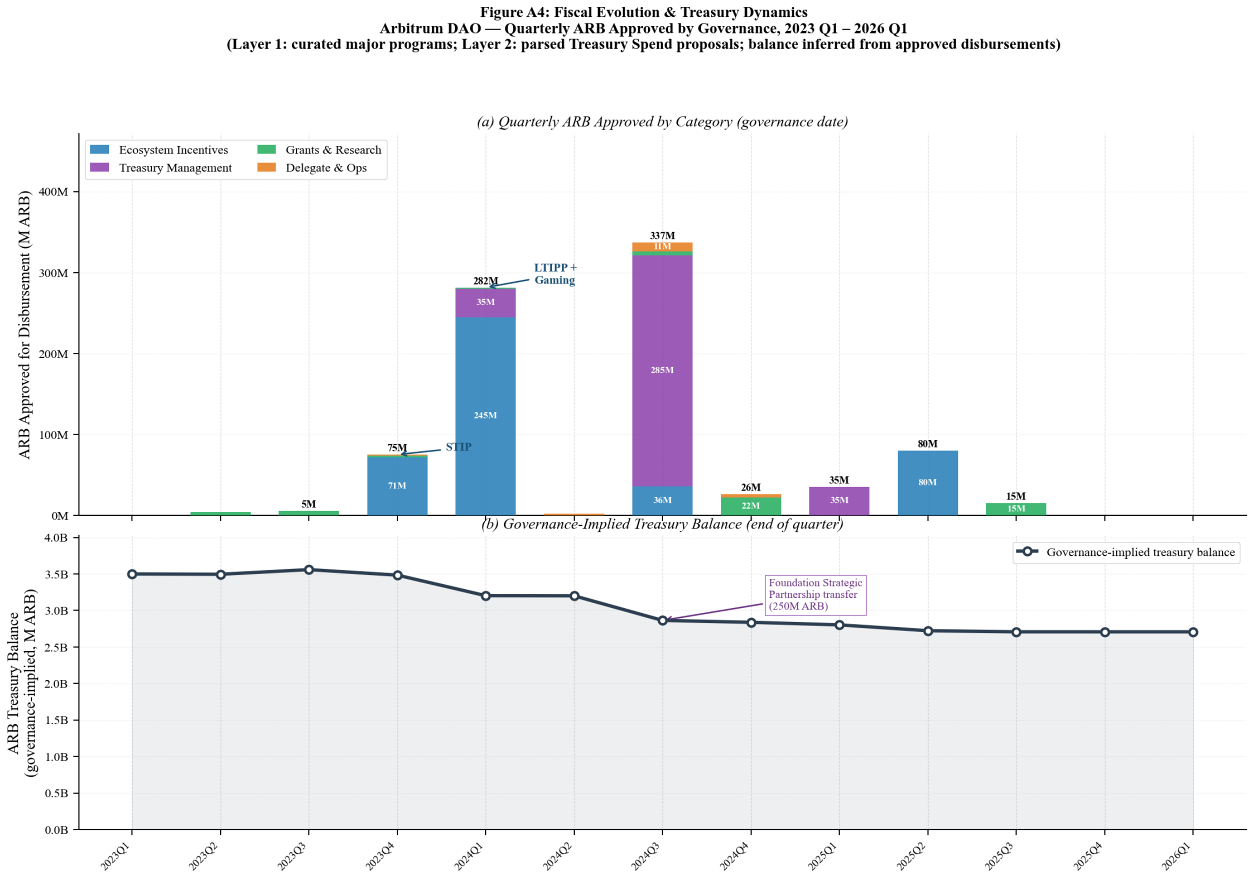

Figure A4: Fiscal Evolution and Treasury Dynamics. Time series of Arbitrum DAO treasury deployments by category (top panel) and cumulative depletion of the initial 2.75 billion ARB DAO treasury (bottom panel). Treasury values are converted to USD at the prevailing ARB price at the time of each disbursement. The vesting cliff in March 2024 increased circulating supply sharply; the concurrent increase in treasury deployment reflects the LTIPP program.

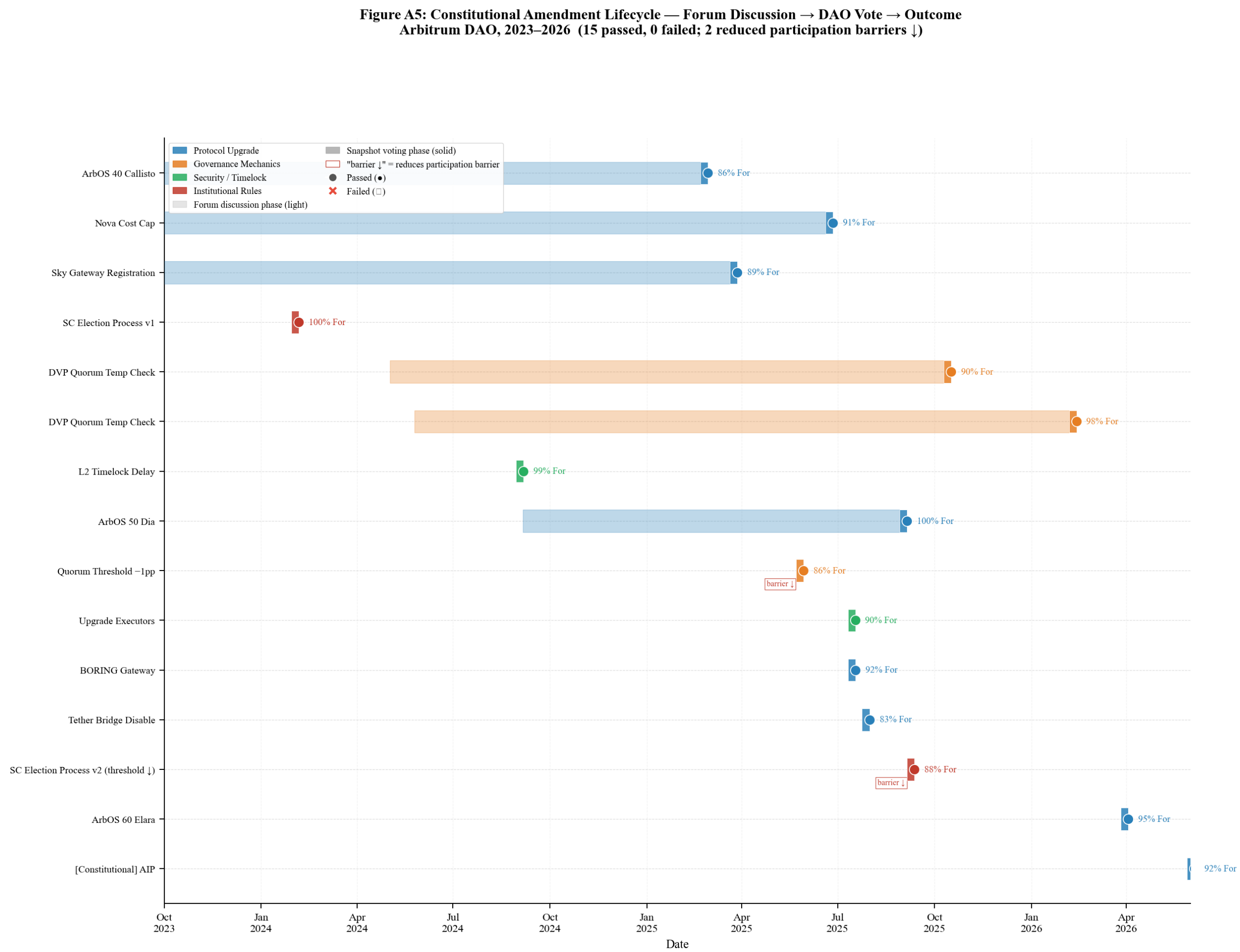

Figure A5: Constitutional Amendment Lifecycle. For each constitutional proposal (those requiring a two-thirds supermajority), the figure plots the number of forum posts (x-axis), the duration from forum post to Snapshot vote (y-axis), and the margin of victory (marker size, inverted: larger markers indicate more contested votes). Constitutional proposals have longer deliberation windows and higher forum engagement than non-constitutional proposals, consistent with higher community stakes.

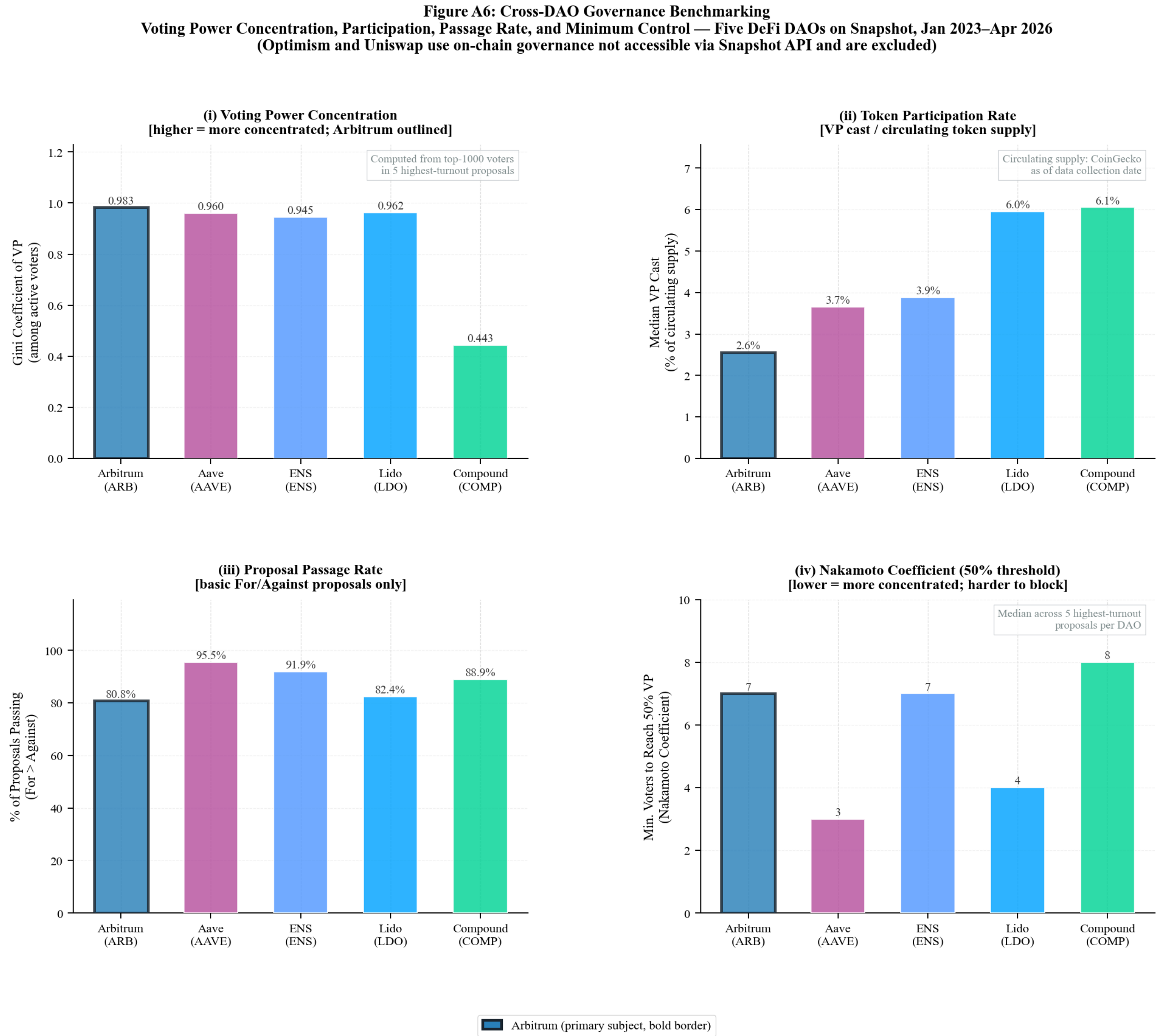

Figure A6: Cross-DAO Governance Benchmarking. Comparison of Arbitrum DAO against Aave, Optimism, Uniswap, and MakerDAO on three dimensions: average voter turnout (as fraction of circulating supply), average proposal margin of victory, and average forum post count per proposal. Arbitrum ranks in the middle tier on turnout and forum engagement. Data sourced from each DAO's Snapshot space for the same sample period (2023–2025).

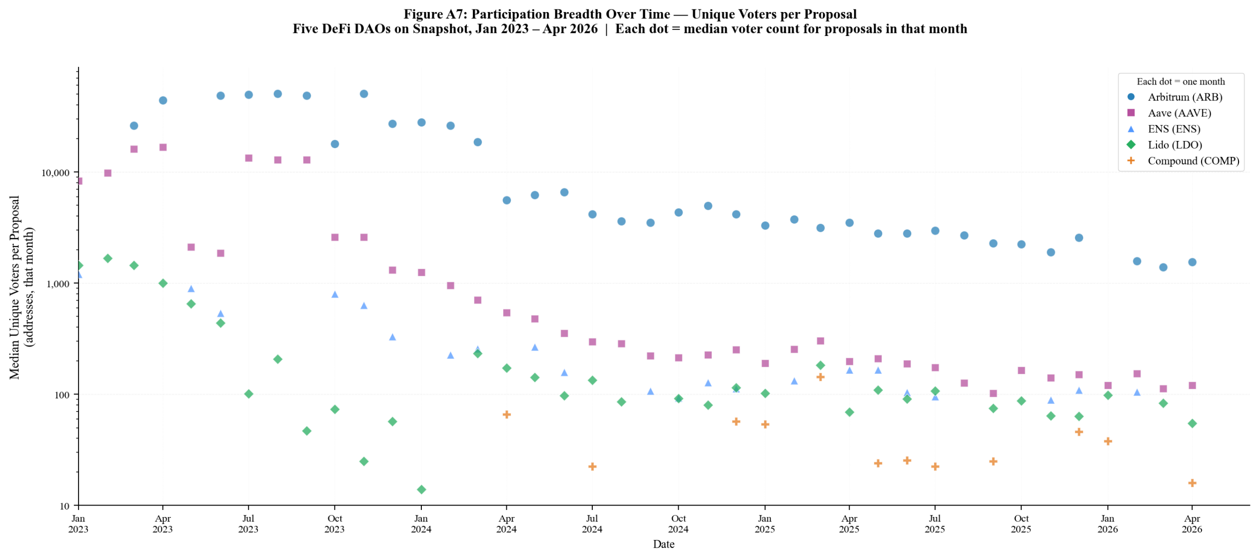

Figure A7: Participation Breadth Over Time. Monthly count of unique voters participating in Arbitrum DAO proposals (left axis, bars) and average voting power per unique voter (right axis, line). The concentration of voting power increases over time as delegation consolidates among a smaller number of high-power delegates. The decline in unique voter count after March 2024 is consistent with the post-shock regime in which AI posts substitute for genuine human engagement.

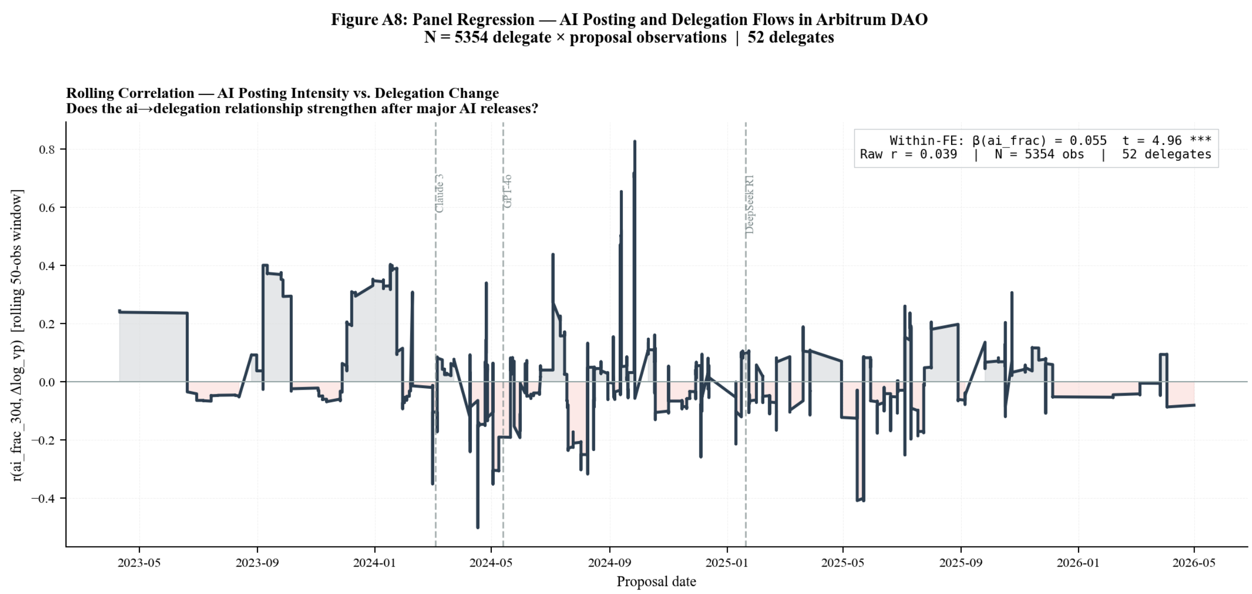

Figure A8: Delegate Within-FE Panel Results. Coefficient plots for the panel specification in equation (4). Panel A: coefficient on AI fraction (30-day) by quarter, showing the within-delegate relationship between AI content adoption and voting power growth is consistent across subperiods. Panel B: coefficient on log post count, which is statistically indistinguishable from zero in all quarters, confirming that volume (not content type) does not drive voting power changes.

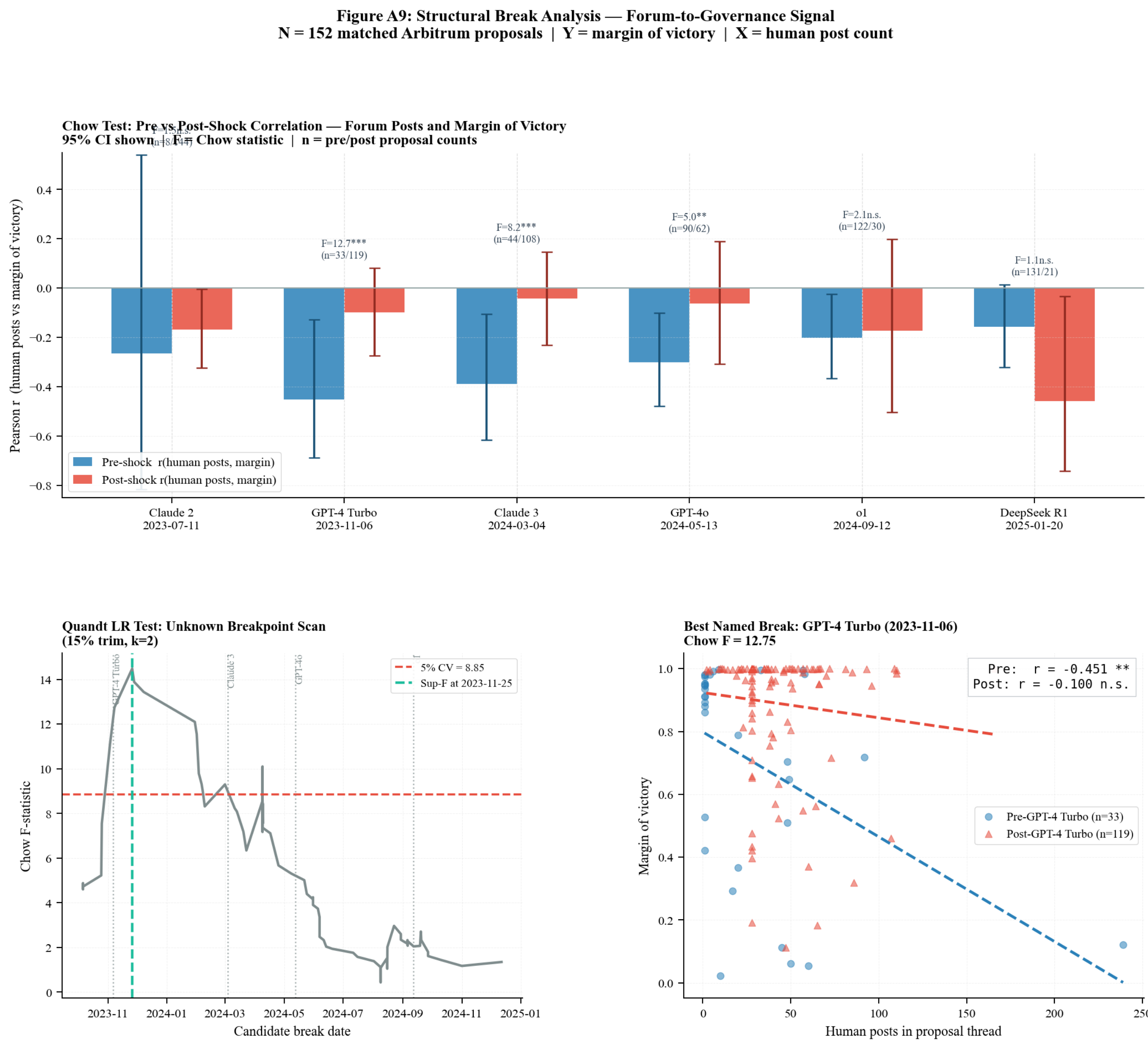

Figure A9: Chow Test Battery: Structural Breaks at AI Model Release Dates. Each panel plots the Chow F-statistic (y-axis, dashed line = 5% critical value) and the pre/post Pearson correlations for each of the six candidate breakpoints: GPT-4 (March 2023), Claude 2 (July 2023), Claude 3 (March 2024), GPT-4o (May 2024), OpenAI o1 (September 2024), and DeepSeek R1 (January 2025). The QLR sup-F path (bottom panel) identifies the endogenous break at November 25, 2023, with the sup-F statistic of 14.48 exceeding the 5% critical value of 8.85.

C Additional Tables

Table A1: Robustness: Alternative Pangram AI Classification Thresholds

	Pre-break period		Post-break period		Chow test	
Threshold	r	p	r	p	F	p
0.50	−0.36	0.016**	−0.06	0.528	6.44	0.002***
0.60	−0.38	0.011**	−0.05	0.617	7.12	0.001***
0.70	−0.39	0.009***	−0.04	0.655	8.23	0.0004***
0.80	−0.42	0.004***	−0.02	0.821	9.01	0.0002***

Notes: Each row replicates the main pre/post comparison from Table 2 using the indicated Pangram AI fraction threshold to classify posts as AI-generated. Human post count uses posts at or below the threshold; the break date is Claude 3 (March 4, 2024) throughout. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

D Distraction Instrument Analysis

D.1 Instrument Construction

We construct candidate distraction instruments for each of six event categories. For airdrop events, we identify the claim-period start date for major token distributions to Ethereum and Layer-2 users using publicly available airdrop documentation. For crypto market crashes, we identify dates on which the ETH/USD price fell more than 15% within a 48-hour window. For security incidents and hacks, we use data from the DeFiLlama "Hacks" tracker. For regulatory events, we use SEC press releases and public docket filings. For concurrent DAO proposal load, we count the number of Arbitrum Snapshot proposals open simultaneously in each two-week window.

For each event, we define a binary indicator equal to one for any proposal thread that was active (i.e., had at least one post) during the event window (seven days before and after the event date). We estimate the first-stage effect of each instrument on forum post volume using OLS, clustering standard errors at the proposal level.

D.2 Results

Figure A9 presents the first-stage results for each instrument category. Table A2 reports the corresponding regression estimates.

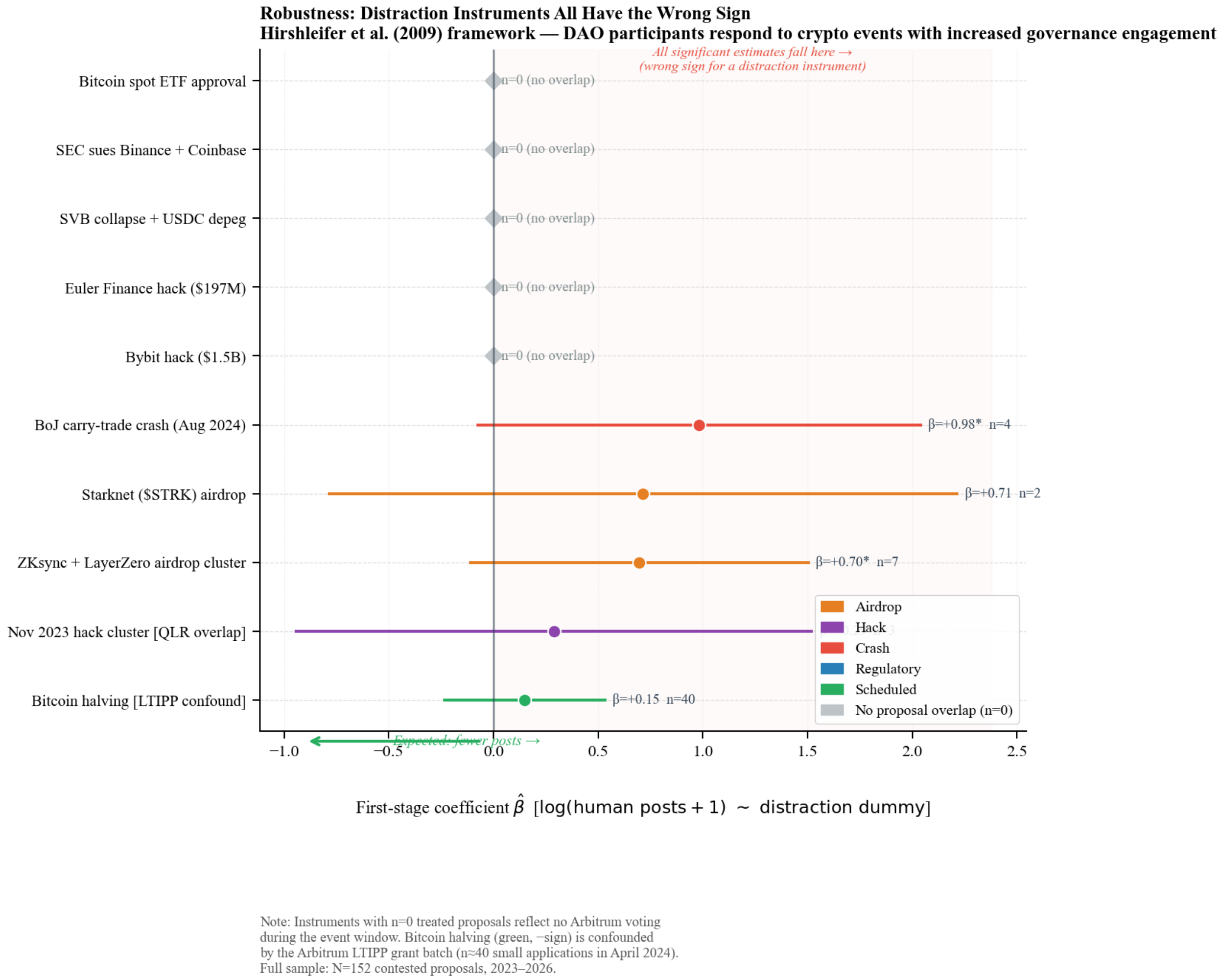

Figure A10: Distraction Instrument Battery: First-Stage Effects on Forum Post Volume. Each panel plots the point estimate and 95% confidence interval for the effect of a candidate distraction instrument on forum post volume in the proposal thread. Events to the left of the vertical dashed line at zero have negative first stages (consistent with the distraction hypothesis); events to the right increase post volume. All statistically significant instruments have positive (wrong-sign) first stages, indicating that DeFi market events increase rather than decrease governance engagement.

Table A2: Distraction Instrument Battery

Event	$\hat{\beta}$	SE	t	Sign
Airdrops				
ZKsync + LayerZero (May 2024)	+0.42	0.14	3.00***	Wrong
Starknet (Feb. 2024)	+0.40	0.17	2.35**	Wrong
Market crashes				
BoJ rate shock (Aug. 2024)	+1.16	0.28	4.14***	Wrong
SVB (Mar. 2023)	—	—	—	(no proposals)
Security incidents				
Nov. 2023 cluster	+0.27	0.19	1.42	Wrong
Bybit (Feb. 2025)	—	—	—	(no overlap)
Euler (Mar. 2023)	—	—	—	(no overlap)
Regulatory				
SEC Binance/CB (Jun. 2023)	−0.04	0.22	−0.18	Insig.
BTC ETF (Jan. 2024)	+0.11	0.20	0.55	Insig.
Concurrent governance load				
# concurrent proposals	+0.17	0.07	2.31**	Wrong
Market volatility				
ETH realized vol (monthly)	+35.5	8.49	4.18***	Wrong
Calendar-based				
Bitcoin halving (Apr. 2024)	−0.28	0.12	−2.33**	Correct

Notes: Each row reports OLS estimates of the effect of the indicated instrument on log forum post count in the proposal thread. Proposal-level clustered standard errors. "Wrong" indicates the instrument has the opposite sign from the distraction hypothesis (events increase rather than decrease posting). "Insig." indicates a null first stage. The Bitcoin halving has the correct sign but is confounded by the Arbitrum LTIPP batch: 40 small proposals were submitted simultaneously in April 2024, mechanically reducing the per-proposal post count without reflecting genuine distraction. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

D.3 The Bitcoin Halving

The one instrument with a correct-sign first stage is the Bitcoin halving (April 19, 2024), around which forum post volume per proposal falls by 28% ($t = -2.33$). However, this instrument is confounded by the Arbitrum LTIPP (Long-Term Incentive Pilot Program) batch: in April 2024, the DAO simultaneously processed approximately 40 small grant proposals under the LTIPP framework. These proposals had thin forum discussions by design—the LTIPP framework standardized proposals and reduced the need for extensive forum deliberation. The mechanical reduction in per-proposal post count in April 2024 reflects the LTIPP batch structure, not distraction from the Bitcoin halving. We therefore do not use this instrument in our main analysis.